

Test Report

Applicant : Makerfabs
6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei
Address : Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China
Product Name : UWB Radio Frequency Module
Brand Mark : 
Model : MaUWB-DW3000 CE
Series model : N/A
Report Number : BLA-EMC-202607-A39-01
Date of Receipt : Jul. 20, 2026
Date of Test : Jul. 20, 2026 to Sep. 07, 2026
Test Standard : ETSI EN 302 065-3-1 V3.2.1 (2025-02)
ETSI EN 303 883-1 V2.1.1 (2024-08)
ETSI EN 303 883-2 V2.1.1 (2024-08)
Test Result : Pass

Compiled by: Mark Chen

Review by: Xavier

Approved by:

Zhang Zheng

Issued Date: Sep. 08, 2026

BlueAsia Technical Services (Shenzhen) Co., Ltd

Address: No.41, South of Beihuan Road, Shangwu Community, Shiyan
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Revise Record

Version No.	Date	Description
01	Sep. 08, 2026	Original

BlueAsia

1 General information

1.1 General information

Applicant	Makerfabs
Address	6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China
Manufacturer	Makerfabs
Address	6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China
Factory	Makerfabs
Address	6th North, Bldg A, ShangHe NanChang Industry Zone, SanWei Community, HangCheng St, Bao'an, ShenZhen, GuangDong, China

1.2 General description of EUT

Product Name	UWB Radio Frequency Module
Model No.	MaUWB-DW3000 CE
Operation Frequency	6.0GHz to 8.5GHz
Test Frequency	6500MHz
Modulation Type	BPM-BPSK
Type of equipment	Type 1
EUT category	1B
Signal characteristic	CASE 4
Number of Channels	1
Antenna Type	Chip antenna
Antenna Gain	5.3dBi(Provided by customer)
Power supply	DC 3.3V
Hardware Version	N/A
Software Version	N/A

Note: For a more detailed description, please refer to Specification or User's Manual supplied by the applicant and/or manufacturer.

2 Test summary

No.	Test item	Reference: Clause No	Result	Remark
1	Operating Frequency Range (OFR)	4.3.2	Pass	
2	Mean e.i.r.p. spectral density	4.3.3	Pass	
3	Peak e.i.r.p. spectral density	4.3.4	Pass	
4	TX unwanted emissions	4.3.5	Pass	
5	Duty Cycle	4.3.6	Pass	Note 1
6	Trigger-before-transmit	4.3.7	Pass	Note 2
7	TX behaviour under the complete environmental profile	4.3.8	Pass	
8	Receiver spurious emissions	4.4.3	Pass	
9	Receiver Baseline Sensitivity (RBS)	4.4.4	Pass	
10	Receiver Baseline Resilience (RBR)	4.4.5	Pass	

N/A: Not Applicable

Note 1: Applies Only applicable to categories 1A, 1B and 2A.

Note 2: Applies Only applicable to categories 1A, and 1B.

2.1 Summary of EUT categories

An overview of requirements for each equipment type and EUT category is given in Table 2.

Table 2: EUT categories according to ECC/DEC/(06)01 [i.4] and Decision (EU) 2024/1467 [i.7]

EUT category	Frequency range	TX requirements		RX-requirements	
		Emission requirements			
			Clause		Clause
1A	3,8 GHz < f ≤ 4,2 GHz or 6,0 GHz < f ≤ 8,5 GHz	OFR	4.3.2	RSE	4.4.3
		Mean e.i.r.p. spectral density	4.3.3	RBS	4.4.4
		Peak e.i.r.p. spectral density	4.3.4	RBR	4.4.5
		TX unwanted emissions	4.3.5		
		Duty Cycle	4.3.6		
		Trigger-before-transmit	4.3.7.2.4		
		TX behaviour in the complete environmental profile	4.3.8		
1B	3,8 GHz < f ≤ 4,2 GHz or 6,0 GHz < f ≤ 8,5 GHz	OFR	4.3.2	RSE	4.4.3
		Mean e.i.r.p. spectral density	4.3.3	RBS	4.4.4
		Peak e.i.r.p. spectral density	4.3.4	RBR	4.4.5
		TX unwanted emissions	4.3.5		
		Duty Cycle	4.3.6		
		Trigger-before-transmit	4.3.7.2.5		
		TX behaviour in the complete environmental profile	4.3.8		
2A	3,8 GHz < f ≤ 4,2 GHz	OFR	4.3.2	RSE	4.4.3
		Mean e.i.r.p. spectral density	4.3.3	RBS	4.4.4
		Peak e.i.r.p. spectral density	4.3.4	RBR	4.4.5
		TX unwanted emissions	4.3.5		
		Duty Cycle	4.3.6		
		TX behaviour in the complete environmental profile	4.3.8		
2B	6,0 GHz < f ≤ 8,5 GHz	OFR	4.3.2	RSE	4.4.3
		Mean e.i.r.p. spectral density	4.3.3	RBS	4.4.4
		Peak e.i.r.p. spectral density	4.3.4	RBR	4.4.5
		TX unwanted emissions	4.3.5		
		TX behaviour in the complete environmental profile	4.3.8		

3 Test Configuration

3.1 Test mode

Test Mode ^{Note 1}	Description
TX	Keep the EUT in continuously transmitting with modulation mode.
RX	Keep the EUT in receiving mode

Note 1: The EUT was configured to measure its highest possible emission and/or immunity level. The test modes were adapted according to the operation manual for use; the EUT was operated in the engineering mode ^{Note 2} to fix the TX or Rx frequency that was for the purpose of the measurements.

Note 2: Special software is used. The software provided by client to enable the EUT under transmission condition continuously at specific channel frequencies individually.

3.2 Operation and Test channel

Channel	Frequency
01	6500MHz

3.3 Test environment

Temperature	Normal	25°C
	Low Extreme	-20°C
	High Extreme	+85°C
Voltage	Normal	DC 3.3V
	Low Extreme	DC 3.2V
	High Extreme	DC 3.4V

3.4 Auxiliary equipment

Device Type	Manufacturer	Model Name	Serial No.	Remark
PC	HP	14s-cr2010TX	/	/
Rechargeable battery	TIANNENG	6-DZF-20.3	/	/

Note:

“--” mean no any auxiliary device during testing.

4 Laboratory information

4.1 Laboratory and accreditations

The test facility is recognized, certified, or accredited by the following organizations:

Company name:	BlueAsia Technical Services (Shenzhen) Co.,Ltd.
Address:	No.41, South of Beihuan Road, Shangwu Community, Shiyao Subdistrict, Bao'an District, Shenzhen, Guangdong, China
Registration No.:	L9788
Telephone:	+86-755-28682673
FAX:	+86-755-28682673

4.2 Measurement uncertainty

This uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of $k=1.96$.

Parameter	Expanded Uncertainty
Radiated Emission(9kHz-30MHz)	$\pm 4.34\text{dB}$
Radiated Emission(30MHz-1000MHz)	$\pm 4.24\text{dB}$
Radiated Emission(1GHz-18GHz)	$\pm 4.68\text{dB}$
Occupied Channel Bandwidth	$\pm 5\%$
RF output power, conducted	$\pm 1.5\text{ dB}$
Power Spectral Density, conducted	$\pm 3.0\text{ dB}$
Unwanted Emissions, conducted	$\pm 3.0\text{ dB}$
Temperature	$\pm 3\text{ }^{\circ}\text{C}$
Supply voltages	$\pm 3\%$
Time	$\pm 5\%$

5 Test equipment

Radiated Spurious Emissions (Below 1GHz)

Equipment	Name	Model	Manufacture	S/N	Cal. Date	Due. Date
BLA-EMC-002-01	Anechoic chamber	9*6*6 chamber	SKET	N/A	2024/3/27	2027/3/26
BLA-EMC-002-02	Control room	966 control room	SKET	N/A	2024/3/27	2027/3/26
BLA-EMC-009	EMI receiver	ESR7	R&S	101199	2025/08/05 2026/07/31	2026/08/04 2027/07/30
BLA-EMC-043	Loop antenna	FMZB1519B	Schwarzbeck	00102	2026/05/31	2027/05/30
BLA-EMC-065	Broadband antenna	VULB9168	Schwarzbeck	01065P	2026/05/31	2027/05/30
BLA-XC-01	Coaxial Cable	N/A	BlueAsia	V01	N/A	N/A
BLA-XC-02	Coaxial Cable	N/A	BlueAsia	V02	N/A	N/A

Radiated Spurious Emissions (Above 1GHz)

Equipment	Name	Model	Manufacture	S/N	Cal. Date	Due. Date
BLA-EMC-001-01	Anechoic chamber	9*6*6 chamber	SKET	N/A	2023/11/16	2026/11/15
BLA-EMC-001-02	Control Room	966 control room	SKET	N/A	2025/11/12	2026/11/11
BLA-EMC-008	Spectrum	FSP40	R&S	100817	2025/08/05 2026/07/31	2026/08/04 2027/07/30
BLA-EMC-013	Horn Antenna	BBHA9120D	Schwarzbeck	01892	2026/05/31	2027/05/30
BLA-EMC-014	Amplifier	PA_000318G-45	SKET	PA201804 3003	2025/08/05 2026/07/31	2026/08/04 2027/07/30
BLA-EMC-046	Filter bank	2.4G/5G Filter bank	SKET	N/A	2026/05/25	2027/05/24
BLA-EMC-061	Receiver	ESPI7	R&S	101477	2026/05/25	2027/05/24
BLA-EMC-066	Amplifier	LNPA_30M01 G-30	SKET	SK202106 0801	2026/05/25	2027/05/24
BLA-EMC-086	Amplifier	LNPA_18G40 G-50dB	SKET	SK202207 1301	2026/05/25	2027/05/24
BLA-EMC-087	Horn Antenna	BBHA 9170	Schwarzbeck	1106	2026/05/31	2027/05/30
BLA-XC-03	Coaxial Cable	N/A	BlueAsia	V03	N/A	N/A
BLA-XC-04	Coaxial Cable	N/A	BlueAsia	V04	N/A	N/A

RF conducted

Equipment	Name	Model	Manufacture	S/N	Cal. Date	Due. Date
BLA-EMC-003-003	Shield room	5*3*3	SKET	N/A	2025/11/12	2026/11/11
BLA-EMC-016	Signal Generator	N5182A	Agilent	MY52420567	2026/05/25	2027/05/24
BLA-EMC-038	Spectrum	N9020A	Agilent	MY49100060	2025/08/05 2026/07/31	2026/08/04 2027/07/30
BLA-EMC-042	Power sensor	RPR3006W	DARE	14I00889SN0 42	2025/08/05	2026/08/04
BLA-EMC-044	Radio communication tester	CMW500	R&S	132429	2025/08/05 2026/07/31	2026/08/04 2027/07/30

BLA-EMC-064	Signal Generator	N5182B	KEYSIGHT	MY58108892	2026/05/25	2027/05/24
BLA-EMC-079	Spectrum	N9020A	Agilent	MY54420161	2025/08/05 2026/07/31	2026/08/04 2027/07/30
BLA-EMC-088	Audio Analyzer	ATS-1	Audio Precision	ATS141094	2026/05/25	2027/05/24

Test Software Record:

Software No.	Software Name	Manufacture	Software version	Test site
BLA-EMC-S001	EZ-EMC	EZ	EEMC-3A1+	RE
BLA-EMC-S002	EZ-EMC	EZ	EEMC-3A1+	RE
BLA-EMC-S010	MTS 8310	MW	2.0.0.0	RF

6 Test result

6.1 Operating Frequency Range (OFR)

The operating frequency range is the frequency range over which the EUT is intentionally transmitting. The operating frequency range of the EUT is determined by the lowest (f_L) and highest frequency (f_H).

For the OFR conformance assessment the number of 10 dB for the parameter X shall be considered (parameter X as specified in ETSI EN 303 883-1

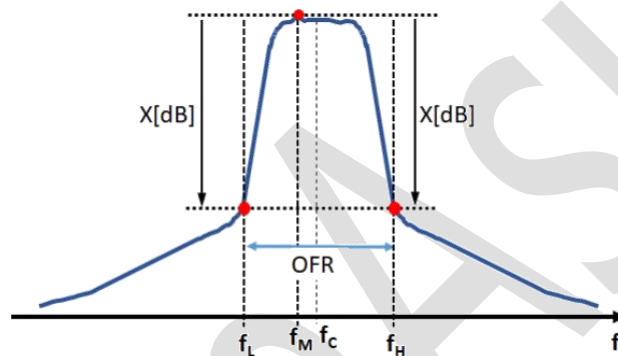


Figure 2a: Operating Frequency Range (OFR)

The lowest (f_L) and highest frequency (f_H) shall be measured based on a specified value X (in [dB]) below the max emission at frequency f_M (see figure 2a). The value for X is given in the related standard. If no value for X is provided in the related standard, then the value of 23 dB shall be used. An EUT can have more than one operating frequency range. In that case each operating frequency range shall be tested separately.

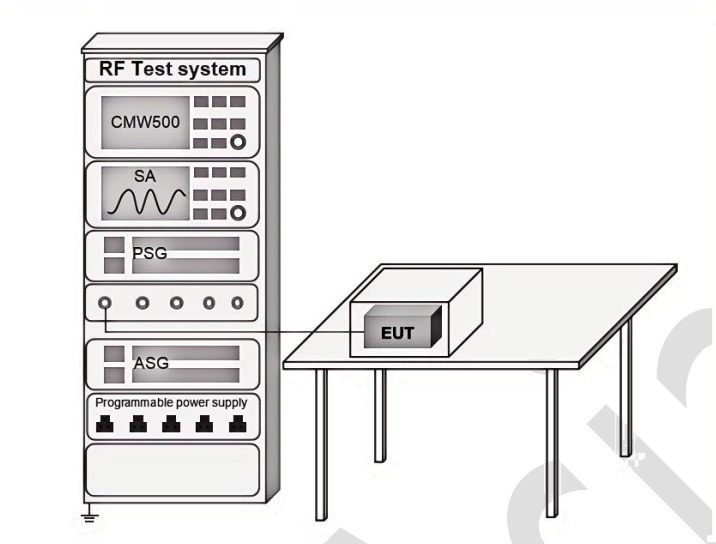
6.1.1 Limit

Any operating bandwidth of all the DUT shall lie within one permitted frequency range of operation of the device (see table 3) and shall be at least 50 MHz.

Table 3: Permitted frequency ranges for vehicular access systems according to ECC/DEC/(06)04 [i.4] and Decision (EU) 2024/1467 [i.7]

	Frequency Range	Application
Transmit and Receive	$3,8 \text{ GHz} < f \leq 4,2 \text{ GHz}$	Vehicular access
Transmit and Receive	$6,0 \text{ GHz} < f \leq 8,5 \text{ GHz}$	Vehicular access

6.1.2 Test setup



6.1.3 Test procedure

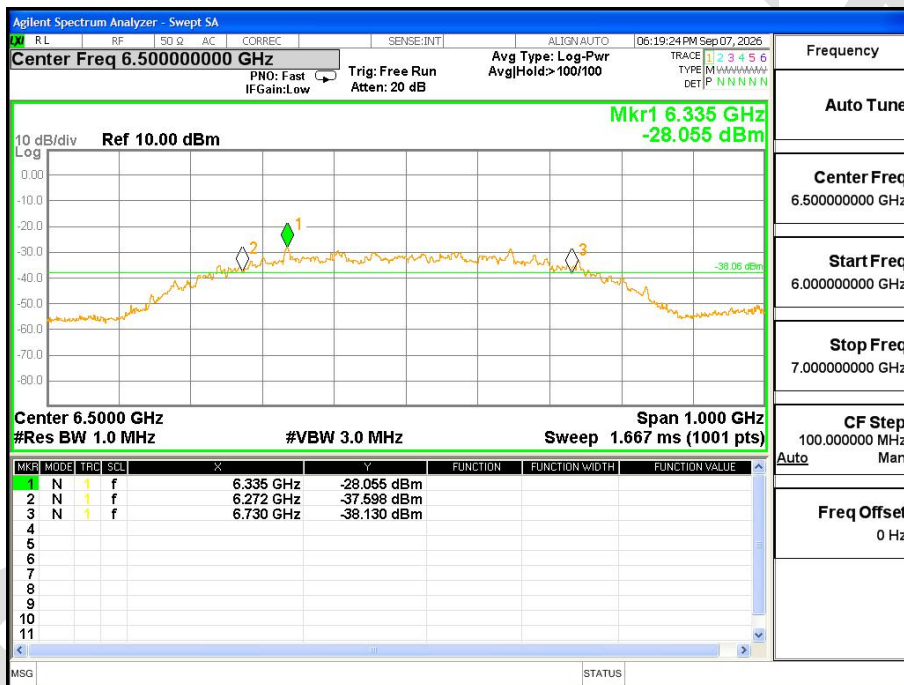
The conformance test for OFR shall be as defined in clause 5.2 of ETSI EN 303 883-1 V2.1.1 (2024-08).

6.1.4 Test data

Test conditions		Test Frequency (MHz)	OFR (MHz)	Limit (MHz)	F _{low} (MHz)	F _{high} (MHz)	Limit (MHz)	Result
Test voltage	Test temperature (°C)							
DC 3.3V	25	6500	458	≥ 50	6272	6730	6000MHz < F _{low} , F _{high} < 8500MHz	Pass

NTNV

-10 dB Bandwidth@6500MHz



6.2 Mean e.i.r.p.spectral density

The mean e.i.r.p. spectral density is defined as the emitted power spectral density over a defined bandwidth of the transmitter including antenna gain radiated in the direction of the maximum level under the specified conditions of measurement.

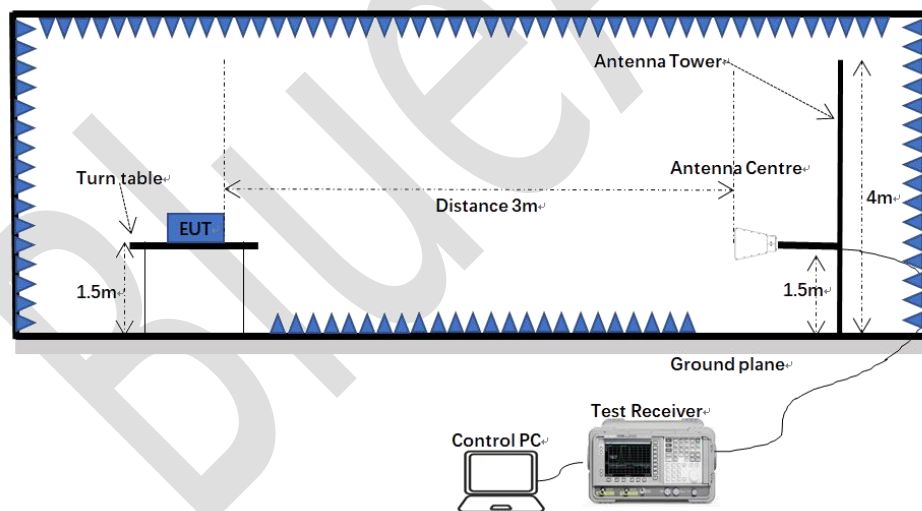
6.2.1 Limit

Within the OFR the Mean e.i.r.p. spectral density shall not exceed the limits in Table 4.

Table 4: Mean e.i.r.p. spectral density limits according to ECC/DEC/(06)04 [i.4] and Decision (EU) 2024/1467 [i.7]

Frequency range	Maximum mean e.i.r.p. spectral density
$3,8 \text{ GHz} < f \leq 4,2 \text{ GHz}$	-41,3 dBm/MHz
$6,0 \text{ GHz} < f \leq 8,5 \text{ GHz}$	-41,3 dBm/MHz

6.2.2 Test setup



(Above 1GHz)

6.2.3 Test procedure

1. Scan from 1GHz to 26GHz; find the maximum radiation frequency to measure.

2. The technique used to find the Spurious Emissions of the transmitter was the antenna substitution method. Substitution method was performed to determine the actual ERP/EIRP emission levels of the EUT.
3. Final Level = Receiver Read level + Correct factor
4. Correct factor = Antenna Factor + Cable Loss – Preamplifier Factor
5. The emission levels of other frequencies are very lower than the limit and not show in test report.

Below 1GHz test procedure as below:

- 1) The EUT was powered on and placed on a table in the chamber. The antenna of the transmitter was extended to its maximum length. Modulation mode and the measuring receiver shall be tuned to the frequency of the transmitter under test.
- 2) Rotating through 360° the turntable. After the fundamental emission was maximized, a field strength measurement was made.
- 3) Steps 1) and 2) were performed with the EUT and the receive antenna in both vertical and horizontal polarization.
- 4) The transmitter was then removed and replaced with another antenna. The center of the antenna was approximately at the same location as the center of the transmitter.
- 5) A signal at the disturbance was fed to the substitution antenna by means of a non-radiating cable. With both the substitution and the receive antennas horizontally polarized, the receive antenna was raised and lowered to obtain a maximum reading at the test receiver. The level of the signal generator was adjusted until the measured field strength level in step 2) is obtained for this set of conditions.
- 6) The output power into the substitution antenna was then measured.
- 7) Steps 5) and 6) were repeated with both antennas vertically polarized.

Above 1GHz test procedure as below:

- 1) Different between above is the test site, change from Semi- Anechoic Chamber to fully Anechoic Chamber.
 - Start frequency: Lower edge of the permitted range of operation (see related standard). If nothing is specified in the related standard the pre-scan method (annex E) should be used.
 - Stop frequency: Upper edge of the permitted range of operation (see related standard). If nothing is specified in the related standard the pre-scan method (annex E) should be used.
 - RBW: 1 MHz.
 - Video Bandwidth: VBW equal or greater than the RBW.
 - Detector Mode: RMS.
 - Display Mode: Max. Hold.
 - Average Time: Equal or larger than signal on-time (T_{on}).
- NOTE: "Average Time" may not be an explicit setting. In some cases, it may be determined by setting the number of measurement points and the time taken for a sweep.
- Number of measurement points: At least equal to frequency span divided by RBW.

- Sweep Time: Time for spectrum analyser sweep (over one Frequency Span) shall be calculated with the following equation (14):

$$\text{Sweep Time (SWT)} \geq \text{signal on-time} \times \text{Number of measurement points} \quad (14)$$

- Signal on- time (T_{on}): See annex C.

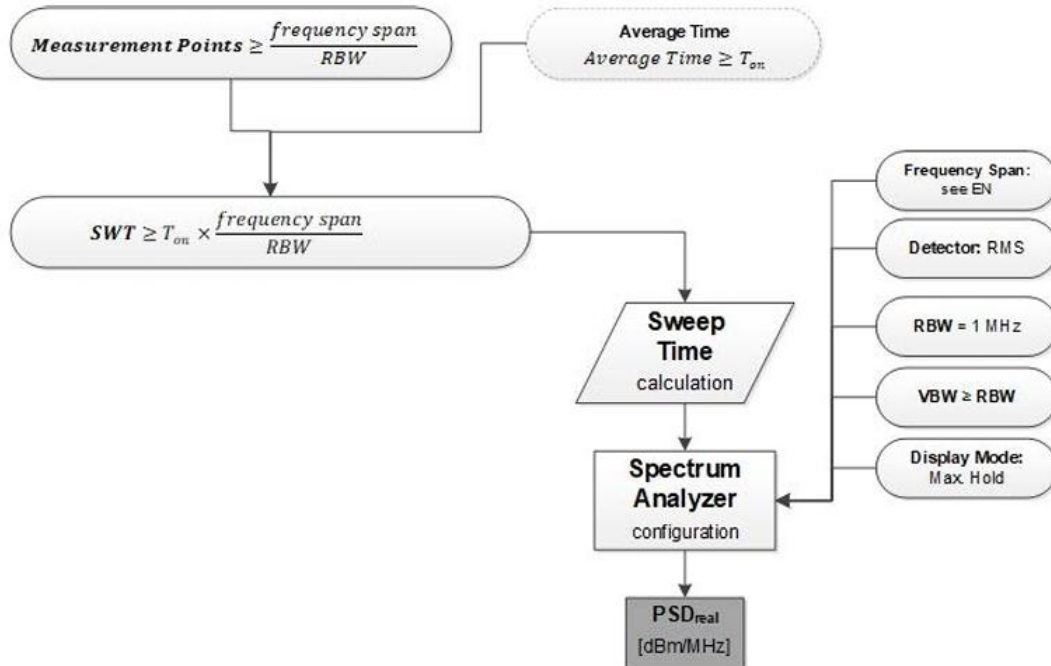


Figure 8: Test Procedure Mean e.i.r.p. spectral density, CASE 4

6.2.4 Test data

Test Frequency (MHz)	Reading Level (dBm)	Correct Factor (dB)	Measurement (dBm/MHz)	Limit (dBm/MHz)	Polarization	Result
6500.0	-62.355	14.20	-48.155	-41.3	Horizontal	PASS
6500.0	-62.948	14.20	-48.748	-41.3	Vertical	PASS

Test conditions		Test Frequency (MHz)	Test value (dBm/MHz)	Limit (dBm/MHz)	Polarization	Result
Test voltage	Test temperature (°C)					
Lowest DC 3.2V	-40	6500.0	-48.526	-41.3	Horizontal	Pass
	-30	6500.0	-48.548	-41.3	Horizontal	Pass
	-20	6500.0	-48.632	-41.3	Horizontal	Pass
	-10	6500.0	-48.699	-41.3	Horizontal	Pass
	0	6500.0	-48.526	-41.3	Horizontal	Pass
	10	6500.0	-48.548	-41.3	Horizontal	Pass
	20	6500.0	-48.632	-41.3	Horizontal	Pass
	30	6500.0	-48.687	-41.3	Horizontal	Pass
	40	6500.0	-48.526	-41.3	Horizontal	Pass
	50	6500.0	-48.548	-41.3	Horizontal	Pass
	60	6500.0	-48.617	-41.3	Horizontal	Pass
	70	6500.0	-48.625	-41.3	Horizontal	Pass
	80	6500.0	-48.514	-41.3	Horizontal	Pass
	85	6500.0	-48.541	-41.3	Horizontal	Pass
Normal DC 3.3V	-40	6500.0	-48.548	-41.3	Horizontal	Pass
	-30	6500.0	-48.632	-41.3	Horizontal	Pass
	-20	6500.0	-48.699	-41.3	Horizontal	Pass
	-10	6500.0	-48.526	-41.3	Horizontal	Pass
	0	6500.0	-48.548	-41.3	Horizontal	Pass
	10	6500.0	-48.155	-41.3	Horizontal	Pass
	20	6500.0	-48.155	-41.3	Horizontal	Pass
	30	6500.0	-48.548	-41.3	Horizontal	Pass
	40	6500.0	-48.632	-41.3	Horizontal	Pass
	50	6500.0	-48.699	-41.3	Horizontal	Pass
	60	6500.0	-48.526	-41.3	Horizontal	Pass
	70	6500.0	-48.548	-41.3	Horizontal	Pass
	80	6500.0	-48.632	-41.3	Horizontal	Pass
	85	6500.0	-48.699	-41.3	Horizontal	Pass
Highest DC 3.4V	-40	6500.0	-48.526	-41.3	Horizontal	Pass
	-30	6500.0	-48.548	-41.3	Horizontal	Pass
	-20	6500.0	-48.632	-41.3	Horizontal	Pass
	-10	6500.0	-48.548	-41.3	Horizontal	Pass
	0	6500.0	-48.632	-41.3	Horizontal	Pass
	10	6500.0	-48.699	-41.3	Horizontal	Pass
	20	6500.0	-48.526	-41.3	Horizontal	Pass
	30	6500.0	-48.542	-41.3	Horizontal	Pass
	40	6500.0	-48.632	-41.3	Horizontal	Pass

	50	6500.0	-48.611	-41.3	Horizontal	Pass
	60	6500.0	-48.369	-41.3	Horizontal	Pass
	70	6500.0	-48.241	-41.3	Horizontal	Pass
	80	6500.0	-48.548	-41.3	Horizontal	Pass
	85	6500.0	-48.632	-41.3	Horizontal	Pass
Lowest DC 3.2V	-40	6500.0	-48.859	-41.3	Vertical	Pass
	-30	6500.0	-48.899	-41.3	Vertical	Pass
	-20	6500.0	-48.963	-41.3	Vertical	Pass
	-10	6500.0	-48.955	-41.3	Vertical	Pass
	0	6500.0	-48.859	-41.3	Vertical	Pass
	10	6500.0	-48.899	-41.3	Vertical	Pass
	20	6500.0	-48.963	-41.3	Vertical	Pass
	30	6500.0	-48.955	-41.3	Vertical	Pass
	40	6500.0	-48.859	-41.3	Vertical	Pass
	50	6500.0	-48.899	-41.3	Vertical	Pass
	60	6500.0	-49.321	-41.3	Vertical	Pass
	70	6500.0	-49.652	-41.3	Vertical	Pass
	80	6500.0	-49.321	-41.3	Vertical	Pass
	85	6500.0	-49.652	-41.3	Vertical	Pass
Normal DC 3.3V	-40	6500.0	-48.859	-41.3	Vertical	Pass
	-30	6500.0	-48.899	-41.3	Vertical	Pass
	-20	6500.0	-49.321	-41.3	Vertical	Pass
	-10	6500.0	-49.652	-41.3	Vertical	Pass
	0	6500.0	-48.859	-41.3	Vertical	Pass
	10	6500.0	-48.748	-41.3	Vertical	Pass
	20	6500.0	-48.748	-41.3	Vertical	Pass
	30	6500.0	-48.878	-41.3	Vertical	Pass
	40	6500.0	-48.891	-41.3	Vertical	Pass
	50	6500.0	-49.321	-41.3	Vertical	Pass
	60	6500.0	-49.652	-41.3	Vertical	Pass
	70	6500.0	-48.963	-41.3	Vertical	Pass
	80	6500.0	-48.899	-41.3	Vertical	Pass
	85	6500.0	-49.321	-41.3	Vertical	Pass
Highest DC 3.4V	-40	6500.0	-49.321	-41.3	Vertical	Pass
	-30	6500.0	-49.652	-41.3	Vertical	Pass
	-20	6500.0	-48.963	-41.3	Vertical	Pass
	-10	6500.0	-48.899	-41.3	Vertical	Pass
	0	6500.0	-49.321	-41.3	Vertical	Pass
	10	6500.0	-49.652	-41.3	Vertical	Pass
	20	6500.0	-48.925	-41.3	Vertical	Pass
	30	6500.0	-49.321	-41.3	Vertical	Pass
	40	6500.0	-49.652	-41.3	Vertical	Pass
	50	6500.0	-48.963	-41.3	Vertical	Pass
	60	6500.0	-48.878	-41.3	Vertical	Pass
	70	6500.0	-49.741	-41.3	Vertical	Pass
	80	6500.0	-49.740	-41.3	Vertical	Pass
	85	6500.0	-48.982	-41.3	Vertical	Pass

6.3 Peak e.i.r.p. spectral density

The maximum peak e.i.r.p. spectral density contained within a 50 MHz bandwidth.

For the peak power spectral density measurements there are two different measurement procedures available. While the method in clause 5.3.4.1.3 applies generally, the method in clause 5.3.4.1.4 applies only if there is only one spectral line in 50 MHz bandwidth at one time and the bandwidth of the spectral line is smaller than the smallest RBW used for the measurements.

In addition the related standard shall specify the test setup (see clause B.1). The conformance test for Peak e.i.r.p. Spectral density (defined within 50 MHz, see clause 5.3.4.1.3 or 5.3.4.1.4) shall be used to measured/asses the emission in each direction around the EUT as specified in the related standard (see test setups in clauses B.1 and B.4)

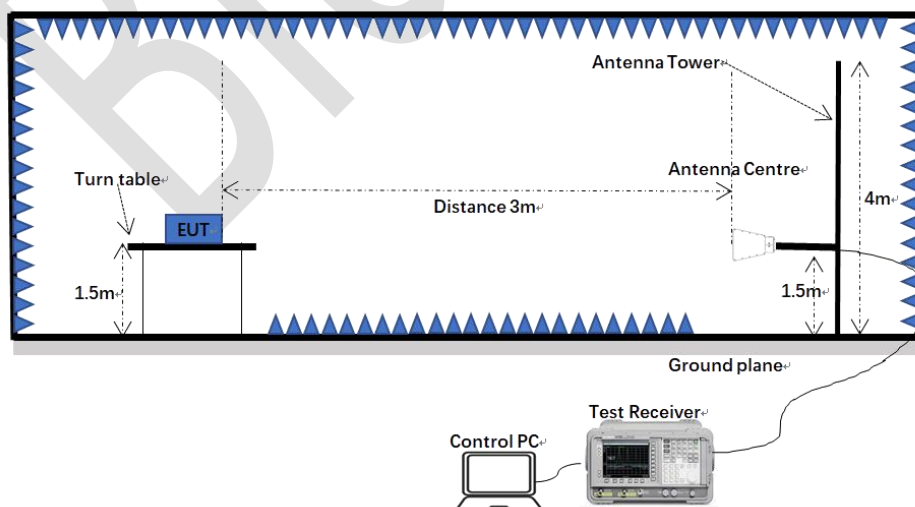
6.3.1 Limit

The Peak e.i.r.p. spectral density shall not exceed the limits in Table 5.

Table 5: Peak e.i.r.p. spectral density limits according to ECC/DEC/(06)04 [i.4] and Decision (EU) 2024/1467 [i.7]

Frequency range	Maximum peak e.i.r.p. (defined in 50 MHz)
$3,8 \text{ GHz} < f \leq 4,2 \text{ GHz}$	0 dBm
$6,0 \text{ GHz} < f \leq 8,5 \text{ GHz}$	0 dBm

6.3.2 Test setup



(Above 1GHz)

6.3.3 Test procedure

6. Scan from 1GHz to 26GHz; find the maximum radiation frequency to measure.
7. The technique used to find the Spurious Emissions of the transmitter was the antenna substitution method. Substitution method was performed to determine the actual ERP/EIRP emission levels of the EUT.
8. Final Level = Receiver Read level + Correct factor
9. Correct factor = Antenna Factor + Cable Loss – Preamplifier Factor
10. The emission levels of other frequencies are very lower than the limit and not show in test report.

Below 1GHz test procedure as below:

- 8) The EUT was powered on and placed on a table in the chamber. The antenna of the transmitter was extended to its maximum length. Modulation mode and the measuring receiver shall be tuned to the frequency of the transmitter under test.
- 9) Rotating through 360° the turntable. After the fundamental emission was maximized, a field strength measurement was made.
- 10) Steps 1) and 2) were performed with the EUT and the receive antenna in both vertical and horizontal polarization.
- 11) The transmitter was then removed and replaced with another antenna. The center of the antenna was approximately at the same location as the center of the transmitter.
- 12) A signal at the disturbance was fed to the substitution antenna by means of a non-radiating cable. With both the substitution and the receive antennas horizontally polarized, the receive antenna was raised and lowered to obtain a maximum reading at the test receiver. The level of the signal generator was adjusted until the measured field strength level in step 2) is obtained for this set of conditions.
- 13) The output power into the substitution antenna was then measured.
- 14) Steps 5) and 6) were repeated with both antennas vertically polarized.

Above 1GHz test procedure as below:

- 1) Different between above is the test site, change from Semi- Anechoic Chamber to fully Anechoic Chamber.

- RBW: Bandwidth equal or greater than 3 MHz and less or equal than 50 MHz.

NOTE 3: It is recommended to use the highest available bandwidth for the signal to be measured because choosing a lower bandwidth could lead to an overestimate of the peak power.

- Video Bandwidth: Not less than the resolution bandwidth.
- Detector Mode: Peak.
- Display Mode: Max. Hold.

Wait for each measurement until the reading in the display is stable (this can be assumed to be achieved if a further increase of time for the max hold measurement will not further change the result anymore).

When the measurements have been performed using a lower bandwidth than 50 MHz, the following correction factor (see equation (16)) shall be added to the measured value before comparison with the peak power limits given in the related standard:

$$\text{Corr}_{\text{dB}} = 20 \times \log_{10} \left(\frac{50 \text{ MHz}}{\text{RBW}_{\text{used}} [\text{MHz}]} \right) \quad (16)$$

The measurement results shall be recorded and shall be assessed against the peak power spectral density limits in the related standard. If the correction based on equation (16) was considered, this shall be recorded in the measurement test report.

6.3.4 Test data

Test Frequency (MHz)	Reading Level (dBm)	Correct Factor (dB)	Measurement (dBm/5MHz)	Polarization	Result
6500	-37.012	14.20	-22.812	Horizontal	PASS
6500	-37.741	14.20	-23.541	Vertical	PASS

Test result:

Test Frequency (MHz)	Measurement Value(dBm/5MHz)	*Corr _{db}	Measurement Value(dBm/50MHz)	Limit (dBm/50MHz)	Result
6500	-22.812	20dB	-2.812	0	Pass

Remark:

Note 1: $*Corr_{db} = 20 \times \log(50\text{MHz} / 5\text{MHz}) = 20\text{dB}$

Note 2: $RBW_{used} [\text{MHz}] = 5\text{MHz}$

Table 4: Spurious emissions limits in line with ERC/REC 74-01 [i.9]

Frequency range	Limit values for TXUE (note)
$87,5 \text{ MHz} \leq f \leq 118 \text{ MHz}$	-54 dBm/100 kHz
$174 \text{ MHz} \leq f \leq 230 \text{ MHz}$	-54 dBm/100 kHz
$470 \text{ MHz} \leq f \leq 694 \text{ MHz}$	-54 dBm/100 kHz
otherwise in band $30 \text{ MHz} \leq f \leq 1\,000 \text{ MHz}$	-36 dBm/100 kHz
$1\,000 \text{ MHz} < f \leq F_{\text{upper}}$ (see table 5)	-30 dBm/1 MHz

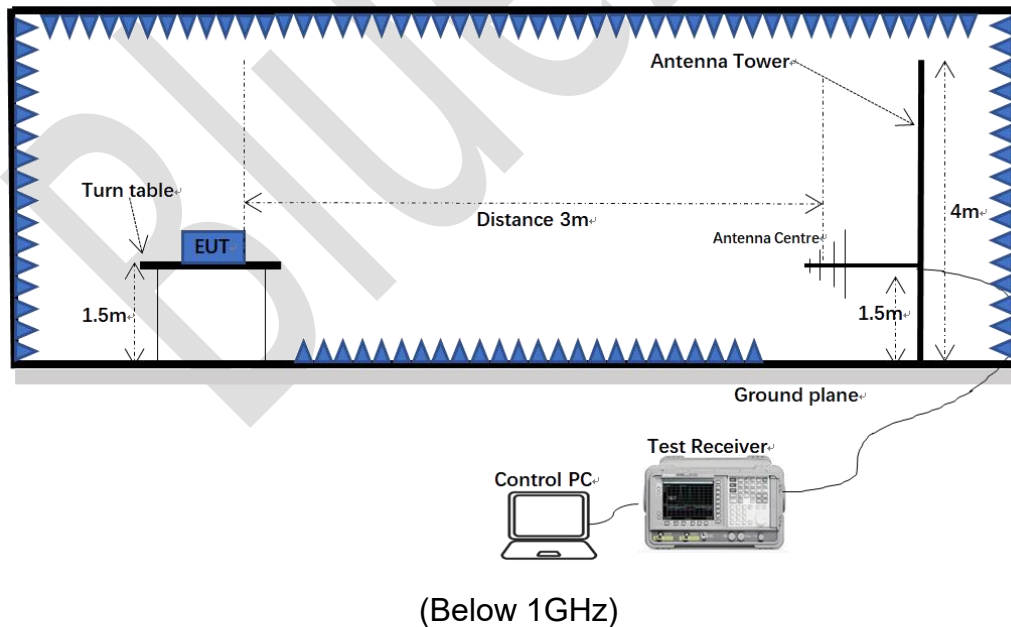
NOTE: Not applicable for RP emissions within the OFR. The limits are mean power averaged over the burst duration

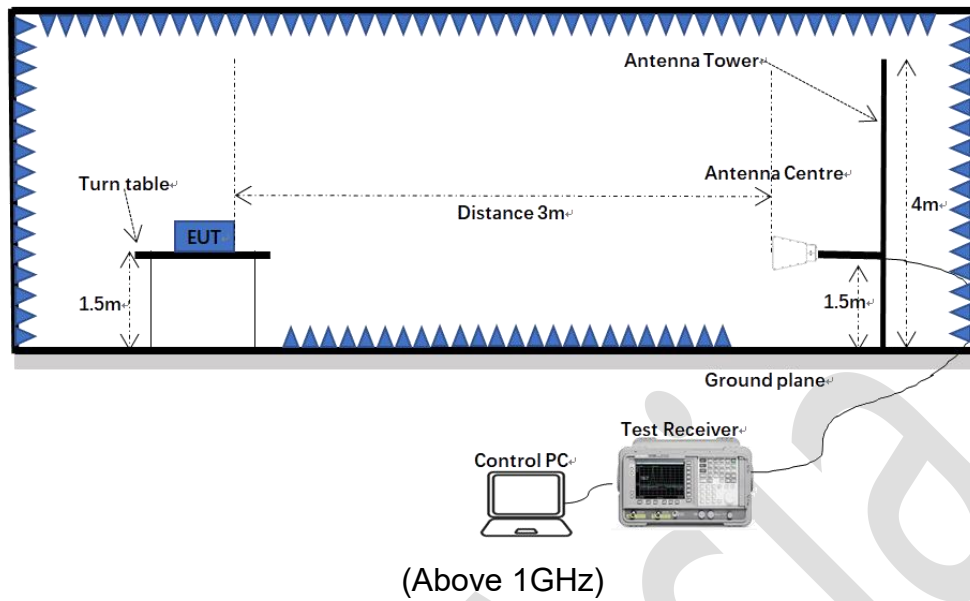
Table 5: Lower and upper frequency for the spurious emissions test based on the EUT OFR in line with ERC/REC 74-01 [i.9]

Fundamental frequency range defined by f_L and f_H (note 2)	Frequency range for measurements	
	Lower frequency (F_{LOWER}) (note 3)	Upper frequency (F_{UPPER})
$300 \text{ MHz} \leq f < 600 \text{ MHz}$	30 MHz	3 GHz
$600 \text{ MHz} \leq f < 5,2 \text{ GHz}$	30 MHz	5 th harmonic (note 1)
$5,2 \leq f < 13 \text{ GHz}$	30 MHz	26 GHz
$13 \leq f < 150 \text{ GHz}$	30 MHz	2 nd harmonic (note 1)
$150 \leq f < 300 \text{ GHz}$	30 MHz	300 GHz

NOTE 1: F_{UPPER} is the stated harmonic of f_H (the upper edge of the OFR, which is measured in clause 5.2).
NOTE 2: F_{LOWER} has to be selected based on f_L and F_{UPPER} based on f_H (f_L and f_H can be measured according to clause 5.2); for receive only devices f_H and f_L of the related EUT/companion device shall be used.
NOTE 3: For EUT operating below 300 MHz the spurious emissions limits below 30 MHz shall be specified in the related standard.

6.4.1 Test setup





6.4.2 Test procedure

11. Scan from 30MHz to 26GHz; find the maximum radiation frequency to measure.
12. The technique used to find the Spurious Emissions of the transmitter was the antenna substitution method. Substitution method was performed to determine the actual ERP/EIRP emission levels of the EUT.
13. Final Level = Receiver Read level + Correct factor
14. Correct factor = Antenna Factor + Cable Loss – Preamplifier Factor
15. The emission levels of other frequencies are very lower than the limit and not show in test report.

Below 1GHz test procedure as below:

- 15) The EUT was powered on and placed on a table in the chamber. The antenna of the transmitter was extended to its maximum length. Modulation mode and the measuring receiver shall be tuned to the frequency of the transmitter under test.
- 16) Rotating through 360° the turntable. After the fundamental emission was maximized, a field strength measurement was made.
- 17) Steps 1) and 2) were performed with the EUT and the receive antenna in both vertical and horizontal polarization.
- 18) The transmitter was then removed and replaced with another antenna. The center of the antenna was approximately at the same location as the center of the transmitter.
- 19) A signal at the disturbance was fed to the substitution antenna by means of a non-radiating cable. With both the substitution and the receive antennas horizontally polarized, the receive antenna was raised and lowered to obtain a maximum reading at the test receiver. The level of the signal generator was adjusted until the measured field strength level in step 2) is obtained for this set of conditions.
- 20) The output power into the substitution antenna was then measured.
- 21) Steps 5) and 6) were repeated with both antennas vertically polarized.

Above 1GHz test procedure as below:

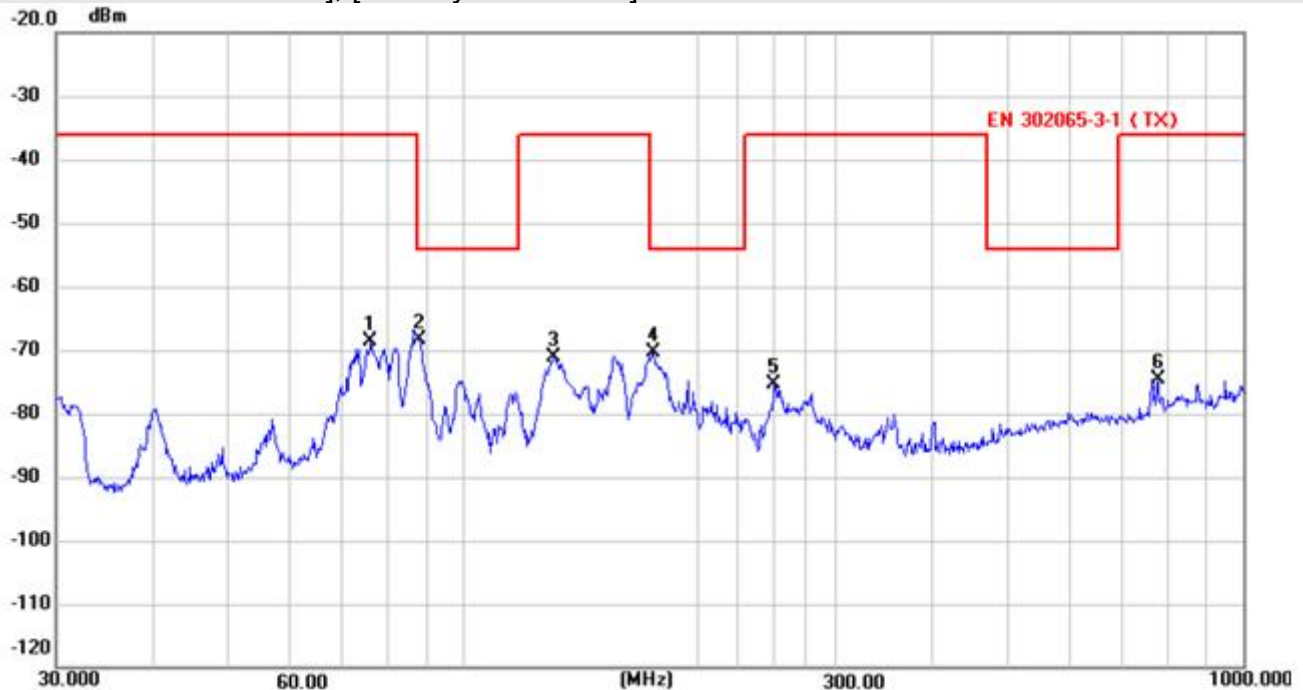
- 1) Different between above is the test site, change from Semi- Anechoic Chamber to fully Anechoic Chamber.

6.4.3 Test data

Note: The points marked on above plots are the highest emissions could be found when testing, so only above points had been displayed. The amplitude of spurious emissions from the radiator which are attenuated more than 20dB below the limit need not be reported, all modes have been tested, and only the worst mode is showed in the report.

6.4.3.1 Below 1GHz

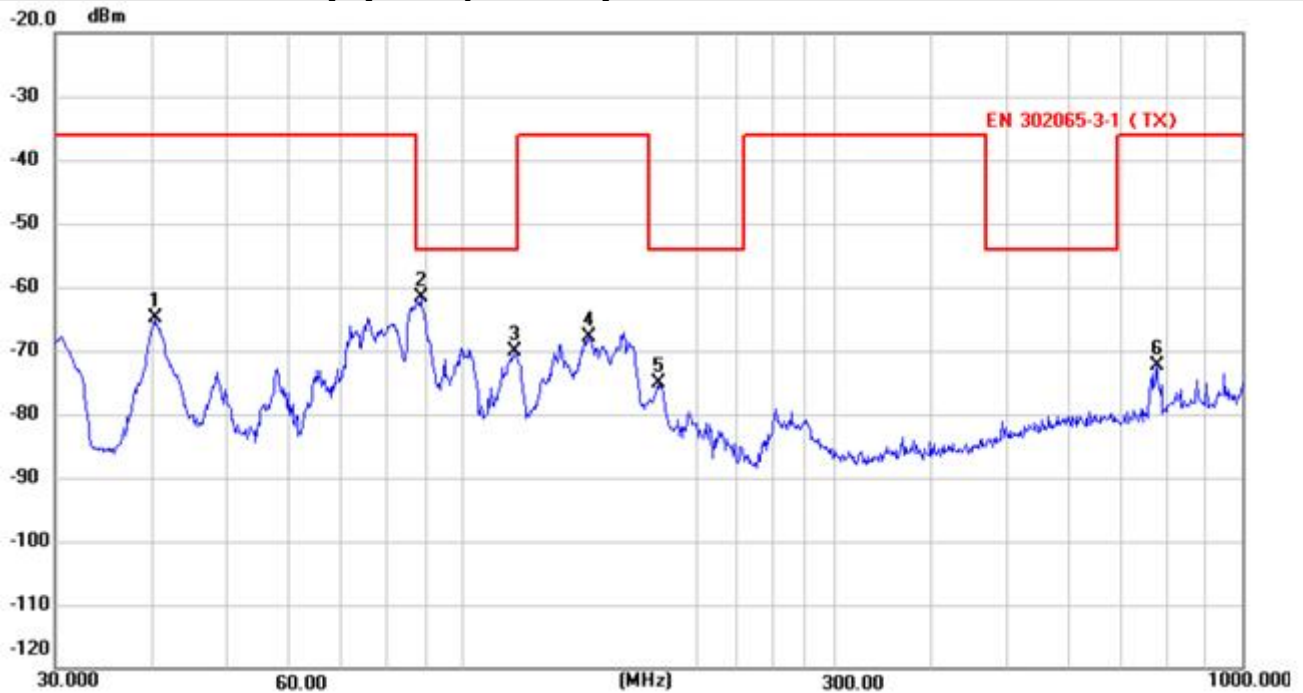
[Test Mode: TX-6500MHz]; [Polarity: Horizontal]



No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector
1	75.9773	-62.73	-5.81	-68.54	-36.00	-32.54	peak
2 *	87.7248	-60.23	-8.26	-68.49	-54.00	-14.49	peak
3	130.3789	-66.74	-4.37	-71.11	-36.00	-35.11	peak
4	175.0368	-65.29	-5.10	-70.39	-54.00	-16.39	peak
5	250.3012	-68.45	-6.90	-75.35	-36.00	-39.35	peak
6	776.8778	-77.50	2.81	-74.69	-36.00	-38.69	peak

Test Result: Pass

[Test Mode: TX-6500MHz];; [Polarity: Vertical]



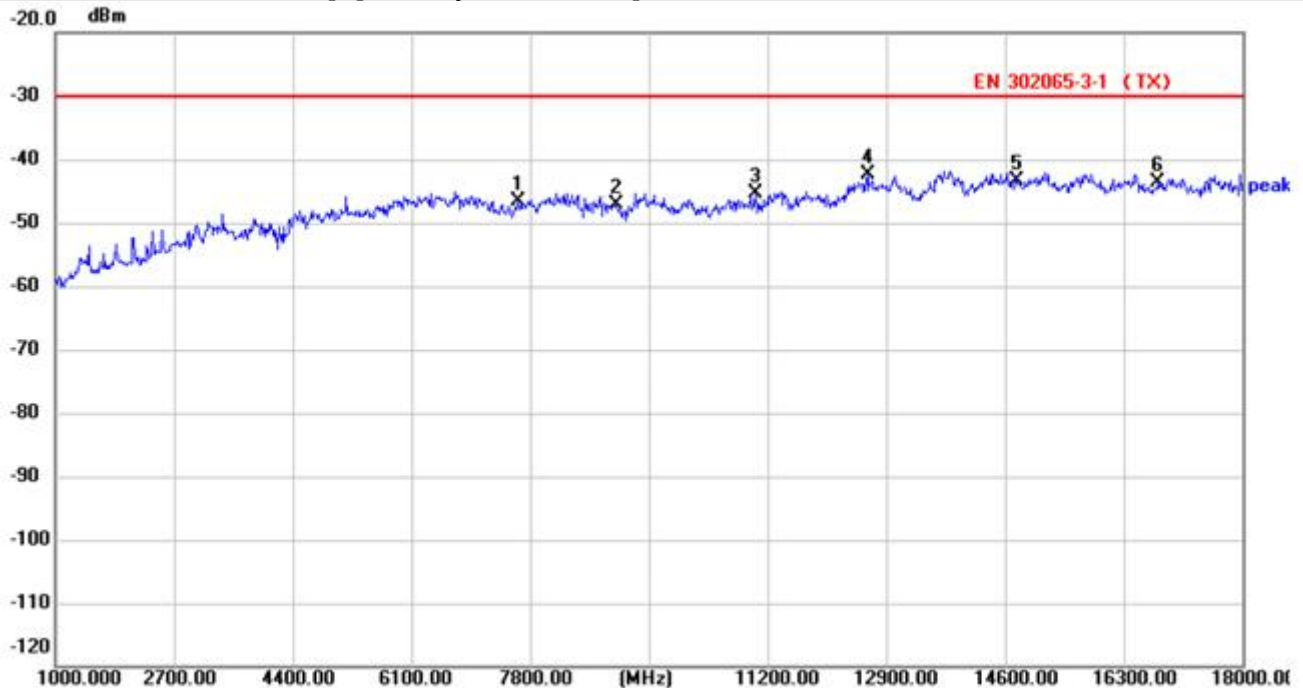
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector
1	40.2757	-55.89	-9.07	-64.96	-36.00	-28.96	peak
2 *	88.3421	-53.36	-8.36	-61.72	-54.00	-7.72	peak
3	116.5401	-63.32	-6.87	-70.19	-54.00	-16.19	peak
4	145.3506	-64.20	-3.59	-67.79	-36.00	-31.79	peak
5	178.1327	-69.58	-5.43	-75.01	-54.00	-21.01	peak
6	776.8778	-75.11	2.81	-72.30	-36.00	-36.30	peak

Test Result: Pass

6.4.3.2 Above 1GHz

Remark: Above 18G Radiated spurious emissions test result more than 20dB below the limit need not be recorded, and only the worst mode is showed in the report.

[Test Mode: TX-6500MHz]; [Polarity: Horizontal]



No.	Mk.	Freq. MHz	Reading Level dBm	Correct Factor dB	Measure- ment dBm	Limit dBm	Over dB	Detector
1		7630.000	-59.84	13.32	-46.52	-30.0	-16.52	peak
2		9024.000	-62.30	15.08	-47.22	-30.0	-17.22	peak
3		11030.00	-62.04	16.79	-45.25	-30.0	-15.25	peak
4	*	12628.00	-62.15	19.67	-42.48	-30.0	-12.48	peak
5		14770.00	-62.03	18.74	-43.29	-30.0	-13.29	peak
6		16793.00	-62.68	18.94	-43.74	-30.0	-13.74	peak

Test Result: Pass

[Test Mode: TX-6500MHz] ; [Polarity: Vertical]



No.	Mk.	Freq. MHz	Reading Level dBm	Correct Factor dB	Measure- ment dBm	Limit dBm	Over dB	Detector
1		6950.000	-59.94	14.26	-45.68	-30.0	-15.68	peak
2		8446.000	-60.63	14.55	-46.08	-30.0	-16.08	peak
3		10724.00	-62.48	16.66	-45.82	-30.0	-15.82	peak
4		12679.00	-63.23	19.89	-43.34	-30.0	-13.34	peak
5	*	13954.00	-61.33	19.13	-42.20	-30.0	-12.20	peak
6		15688.00	-61.16	18.82	-42.34	-30.0	-12.34	peak

Test Result: Pass

6.5 Duty Cycle

Short-Term Duty Cycle For the Short-Term Duty Cycle conformance test the set-up as specified in clause 5.3.3 and the conformance test procedure as specified in ETSI EN 303 883-1 [1], clause 5.11.2.1 ("Duty cycle, spectrum analyser method") shall be used.

The measurement receiver with the related measurement antenna shall be placed in the direction of highest Mean e.i.r.p spectral density emission (see clause 5.3.3).

NOTE 1: Duty Cycle conformance assessment is only necessary in the direction of the highest Mean e.i.r.p spectral density emission, because with this test the nature of the TX-signal will be tested and not the complete emission of the EUT (see clause 5.3.3).

For the measurement distance d , the same distance than for the Mean e.i.r.p spectral density emission conformance test shall be used (see clause 5.3.3).

If based on the low emission levels of the EUT the noise level of the overall measurement system (ETSI EN 303 883-1 [1], clause B.2.5) has less than 16 dB margin to the measured $P_{\max}$ (see below), a smaller measurement distance d shall be used, and the measurement distance assessment as described as range length in ETSI EN 303 883-1 [1], clause B.2.3.5 shall be done.

Following parameters shall be used for Short-Term-Duty-Cycle tests:

- $T_{\text{obs}} = 1 \text{ s}$
- $T_{\text{dis}} = 10 \mu\text{s}$

NOTE 2: This value for disregard time T_{dis} is large enough to span pulse-based symbols and include them in the On-time, but small enough to differentiate packets and account the time between as Off-time.

- $\text{RBW} = 1 \text{ MHz}$
- $\text{VBW} = 3 \text{ MHz}$
- Detector mode: Peak
- $P_{\text{thresh}} = P_{\text{max}} - 10 \text{ dB}$

NOTE 3: P_{thresh} and P_{max} are related to the duty-cycle measurement settings ($\text{RBW} = 1 \text{ MHz}$, Peak Detector). P_{max} denotes the maximum level of the signal measured.

6.5.1 Limit

The measured Duty Cycle shall comply with the limits in Table 8.

Table 8: Duty Cycle limits according to ECC/DEC/(06)04 [i.4] and Decision (EU) 2024/1467 [i.7]

Limits	Remarks
$T_{on\ max} = 5\ ms$ $T_{off\ mean} \geq 38\ ms$ (averaged over 1 s) $\Sigma\ T_{off} > 950\ ms\ per\ s$	Short-term duty-cycle requirements for $T_{obs} = 1\ s$: $\Sigma\ T_{off} > 950\ ms$ sets the duty-cycle limit to 5 % (see notes 2 and 3) $T_{off\ mean} \geq 38\ ms$ implicitly limits the number of On-Periods and thus the achievable duty-cycle for short T_{on} times
$\Sigma\ T_{on} < 18\ s\ per\ hour$	Long-term duty-cycle requirement for $T_{obs} = 3\ 600\ s$ $\Sigma\ T_{on} < 18\ s$ sets the duty-cycle limit per hour to 0,5 %
NOTE 1: The limits are based on the "Low Duty Cycle" requirements in ECC/DEC/(06)04 [i.4], annex 2. NOTE 2: 5 % Duty-Cycle can only be achieved with T_{on} times larger than 2 ms (and maximum 5 ms). NOTE 3: The limits allow a Short-Term Duty-Cycle of up to 5 %. Devices with $T_{on} < 2\ ms$ cannot exploit this Short-Term Duty-Cycle of 5 %, as $T_{off\ mean}$ is the limiting factor. The resulting Short-Term Duty-Cycle for those devices is then considerably smaller than 5 %.	

6.5.2 Test data

The manufacturer has declared that due to the time between transmissions by the devices, the duty cycle $\leq 5\%$.

Conclusion: Pass

For the Long-Term Duty Cycle conformance test the set-up as specified in clause 5.3.3 and the conformance test procedure as specified in ETSI EN 303 883-1 [1], clause 5.11.2.1 ("Duty cycle, spectrum analyser method") shall be used.

The measurement receiver with the related measurement antenna shall be placed in the direction of highest Mean e.i.r.p spectral density emission (see clause 5.3.3).

NOTE 1: Duty Cycle conformance assessment is only necessary in the direction of the highest Mean e.i.r.p spectral density emission, because with this test the nature of the TX-signal will be tested and not the complete emission of the EUT (see clause 5.3.3).

For the measurement distance d, the same distance than for the Mean e.i.r.p spectral density emission conformance test shall be used (see clause 5.3.3).

If based on the low emission levels of the EUT the noise level of the overall measurement system (ETSI EN 303 883-1 [1], clause B.2.5) has less than 16 dB margin to the measured P_{max} (see below), a smaller measurement distance d shall be used, and the measurement distance assessment as described as range length in ETSI EN 303 883-1 [1], clause B.2.3.5 shall be done.

Following parameters shall be used for Long-Term-Duty-Cycle assessment:

- $T_{\text{obs}} = 60 \text{ s}$
- $T_{\text{dis}} = 10 \text{ }\mu\text{s}$

NOTE 2: This disregard time is large enough to span pulse-based symbols and include them in the On-time, but small enough to differentiate packets and account the time between as Off-time.

- $\text{RBW} = 1 \text{ MHz}$
- $\text{VBW} = 3 \text{ MHz}$
- Detector mode: Peak
- $P_{\text{thresh}} = P_{\text{max}} - 10 \text{ dB}$

The measured ΣT_{on} in 60 s shall be used to extrapolate the cumulated on-time for 1 hour:

$$\Sigma T_{\text{on}} (1 \text{ hour}) = \Sigma T_{\text{on}} (60 \text{ s}) \times 60$$

NOTE 3: The Long-Term-Duty-Cycle requirement is implicitly fulfilled, if the EUT is compliant to "Trigger-before-transmit".

NOTE 4: The extrapolation from 60 s to 1 hour does not put a harder Long-Term-Duty-Cycle requirement to the EUT, if the T_{on} time of one transmission is less than 190 μs (because the 38 ms T_{off} mean requirement is the limiting parameter). This is typically the case for Keyless Entry protocols.

6.5.3 Test data

The manufacturer has declared that due to the time between transmissions by the devices, the duty cycle $\leq 0.5\%$.

Conclusion: Pass

BlueAsia

6.6 Trigger-before-transmit

For the Trigger-before-transmit conformance test the set-up as specified in clause 5.3.3 shall be used. The companion device shall be placed at 1 m distance in the direction of the highest Mean e.i.r.p spectral density emission (see clause 5.3.3).

Conformance for "Trigger-before-transmit" shall be tested using a procedure as below.

The measurement can be radiated or conducted, using a spectrum analyser in Zero-Span with settings as for clause 4.3.6 ("Duty Cycle") or an oscilloscope - those methods provide equivalent results for the time domain.

Depending on the trigger scenario one of following test setups and procedures shall be used:

- Trigger Scenario 1: see clause 5.3.7.2.
- Trigger Scenario 2: see clause 5.3.7.3.

5.3.7.2 Setup for Trigger Scenario 1

Figure 7 shows the setup for Trigger Scenario 1, where EUT is responder, and trigger is a received UWB packet.

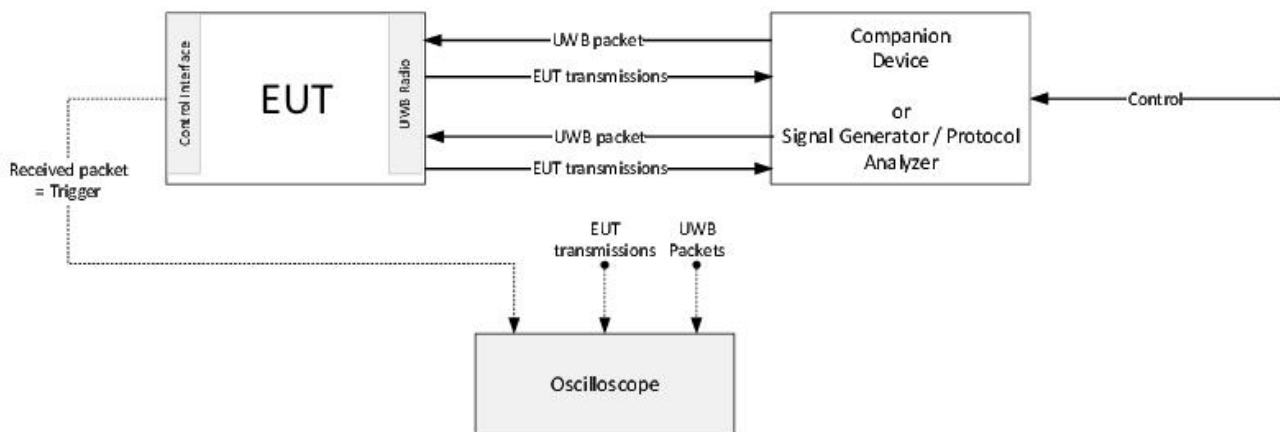


Figure 7: Setup for Trigger Scenario 1

The manufacturer shall provide a companion device including control equipment or provide necessary information how to configure signal generator and protocol analyser. (In the following text, "companion device" will be used for either embodiment.)

Procedure:

- 1) Companion device sends a single packet:
 - a) Verify at EUT control interface that packet has been received.
 - b) Measure TBT timeout and cumulated ON-time at EUT UWB radio interface, relative to the sent packet from the companion device.
 - c) The measured values shall comply with the Trigger-before-transmit limits.

- 2) Companion device sends repeating packets, repetition time < 10 s:
 - a) Measure cumulated ON-time T_{cum_on} within any 10 s.
 - b) Vary (reduce) repetition time, to determine the worst case cumulated ON-time T_{cum_on} .
 - c) The maximum value for T_{cum_on} shall be recorded and shall comply with the Trigger-before-transmit limit.
- 3) Deactivate repeating packet transmissions:
 - a) Verify that EUT stops transmitting.
 - b) Measure TBT timeout, relative to the last received UWB packet.
 - c) The larger value of 1b or 2b shall be recorded as result for $TBT_{timeout}$ and shall comply with the limit.

The test procedure shall include at least 10 repetitions of the trigger signal, before deactivating the companion device. The repetitions shall confirm identical behaviour of the EUT or reveal worst-case behaviour.

The observation time after the (latest) trigger shall be at least 1 min to confirm that the EUT has ceased UWB transmissions.

The test shall cover all operational modes of the EUT.

5.3.7.3 Setup for Trigger Scenario 2

Figure 8 shows the setup for Trigger Scenario 2, where EUT is initiator, and trigger is a control message.

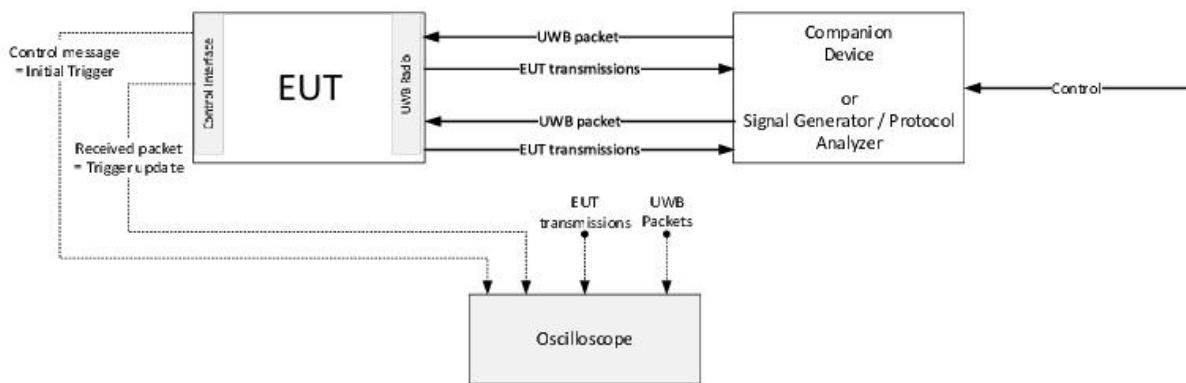


Figure 8: Setup for Trigger Scenario 2

The manufacturer shall provide a companion device including control equipment or provide the necessary information how to configure signal generator and protocol analyser. (In the following text, "companion device" will be used for either embodiment.)

Procedure:

- 1) Control equipment sends a control message to EUT:
 - a) EUT sends one or more transmissions.
 - b) Measure TBT timeout and cumulated ON-time at EUT UWB radio interface, relative to the start of the first EUT transmission.
 - c) The measured values shall comply with the Trigger-before-transmit limits.
- 2) Control equipment sends two control messages within cease time to EUT:
 - a) The second control message shall be sent still within the cease time of the first control message (=EUT trigger).
 - b) Measure TBT timeout and cumulated ON-time at EUT UWB radio interface, relative to the start of the first EUT transmission.
 - c) The second control message shall be ignored and shall not lead to further transmissions by the EUT. The measured values shall comply with the Trigger-before-transmit limit and are expected to be the same as in 1).
- 3) Companion device sends response packets:
 - a) Control equipment sends control message to EUT as initial trigger.
 - b) EUT sends one or more transmissions.
 - c) The companion device shall send responses to the EUT.
 - d) Verify at EUT control interface that response packets are received.
 - e) Measure cumulated ON-time T_{cum_on} within any 10 s.
 - f) The maximum value for T_{cum_on} shall be recorded and shall comply with the Trigger-before-transmit limit.
- 4) Deactivate response packet transmissions:
 - a) Verify that EUT stops transmitting.
 - b) Measure TBT timeout, relative to the last received UWB packet.
 - c) The larger value of 1b or 4b shall be recorded as result for $TBT_{timeout}$ and shall comply with the limit.

The test procedure shall include at least 10 response packets, before deactivating the companion device. Evaluation of at least 10 response packets shall confirm identical behaviour of the EUT or reveal worst-case behaviour.

The observation time after the (latest) trigger shall be at least 1 min to confirm that the EUT has ceased UWB transmissions.

6.6.1 Limit

The limits given in Table 9 shall not be exceeded.

Table 9: Trigger-before-transmit Limits

Trigger-Before-Transmit timeout $TBT_{timeout}$ Cease time for UWB transmissions after EUT trigger (reception of UWB packets resets the timeout)	max 10 s
Trigger-Before-Transmit On-Time $TBT_{On-Time}$ Cumulated Ton time T_{on_cum} within any 10 s window after any trigger event	max 50 ms

Figure 4 is a graphical interpretation of the Trigger-before-transmit limits.

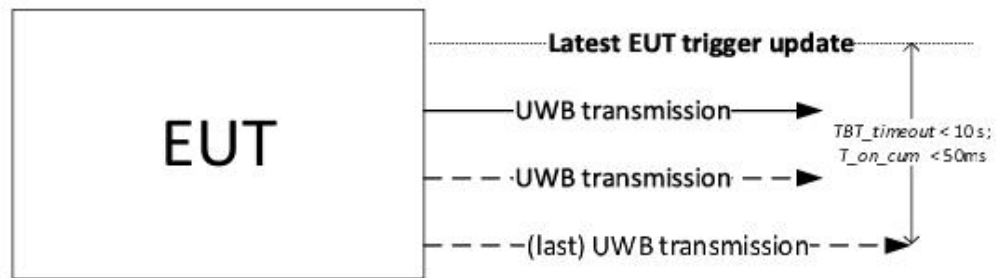


Figure 4: Trigger-before-transmit limits

For the justification of Trigger-before-transmit limits see annex C.

6.6.2 Test data

The manufacturer has declared that due to the time between transmissions by the devices, Trigger-Before-Transmit timeout $TBT_{timeout} < 10s$. Trigger-Before-Transmit On-Time $TBT_{On-Time} < 50ms$.

Conclusion: Pass

6.7 TX behaviour under the complete environmental profile

1. The TX behaviour under the complete environmental profile requirement applies to all EUT categories as described in Table 2 in clause 4.2.3.
2. The TX behaviour under the complete environmental profile verifies the conformance of the Mean e.i.r.p. spectral density over the environmental profile as specified in clause 5.1.3.
The complete environmental profile test conditions cover both the normal (see clause 5.1.2) and extreme test conditions.
3. The temperature range specified for the equipment type shall cover the temperature range of Table 14.
4. The complete environmental profile conditions shall be as defined in clause ETSI TS 103 941 [3], clause 4.5.3.2 with the temperature range of Table 14:

Table 14: Temperature range for the complete environmental profile test

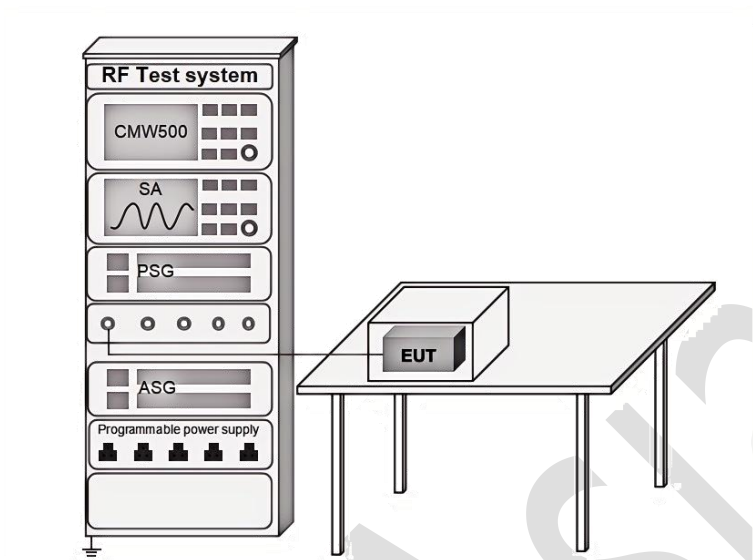
Type of equipment	Minimum	Maximum
Type 1	-40 °C	+85 °C
Type 2	0 °C	+50 °C

The nominal supply voltage shall be as specified for the intended use for the equipment (e.g. in the user manual or technical specification file)

6.7.1 Limit

The TX behaviour is obtained by measuring the Mean e.i.r.p. spectral density across the complete environmental profile (as specified in clause 5.1.3 and assessing the variation with respect to an adjusted Regulated Limit (RL)). The procedure to adjust the regulated limit is described in ETSI TS 103 941 [3], clause 6.4.

6.7.2 Test setup



6.7.3 Test procedure

A test set-up shall be chosen based on clause 4.3.1 and Figure 1 in ETSI TS 103 941 [3].

The procedure according to ETSI TS 103 941 [3], clause 6.4 shall be used.

Based on clause 5.1.3 and Figure 6 of ETSI TS 103 941 [3], clause 4.5.4 the parameters for the assessment are specified

as follow:

- tlow: see clause 5.1.3, Table 14 of ETSI EN 302 065-3-1 V3.2.1 (2025-02)
- thigh: see clause 5.1.3, Table 14 of ETSI EN 302 065-3-1 V3.2.1 (2025-02)
- tsteps: 10 °C
- supply voltage: see clause 5.1.3 of ETSI EN 302 065-3-1 V3.2.1 (2025-02), the nominal value of the supply voltage is usually provided by the user manual of the EUT.

6.7.4 Test data

Pass

Note: The TX behaviour under the complete environmental profile has been reported on the project OFR and Mean e.i.r.p spectral density, meeting the requirements of the test

6.8 Receiver spurious emissions

The Receiver spurious emissions requirement applies to all EUT categories as described in Table 2 in clause 4.2.3 of of ETSI EN 302 065-3-1 V3.2.1 (2025-02)

For the description of the Receiver spurious emissions, see ETSI EN 303 883-2 [2], clause 5.2.1.

6.8.1 Limit

The Receiver spurious emissions shall comply with limits in Table 10.

Table 10: Receiver spurious emission limits according to ERC Recommendation 74-01 [i.15]

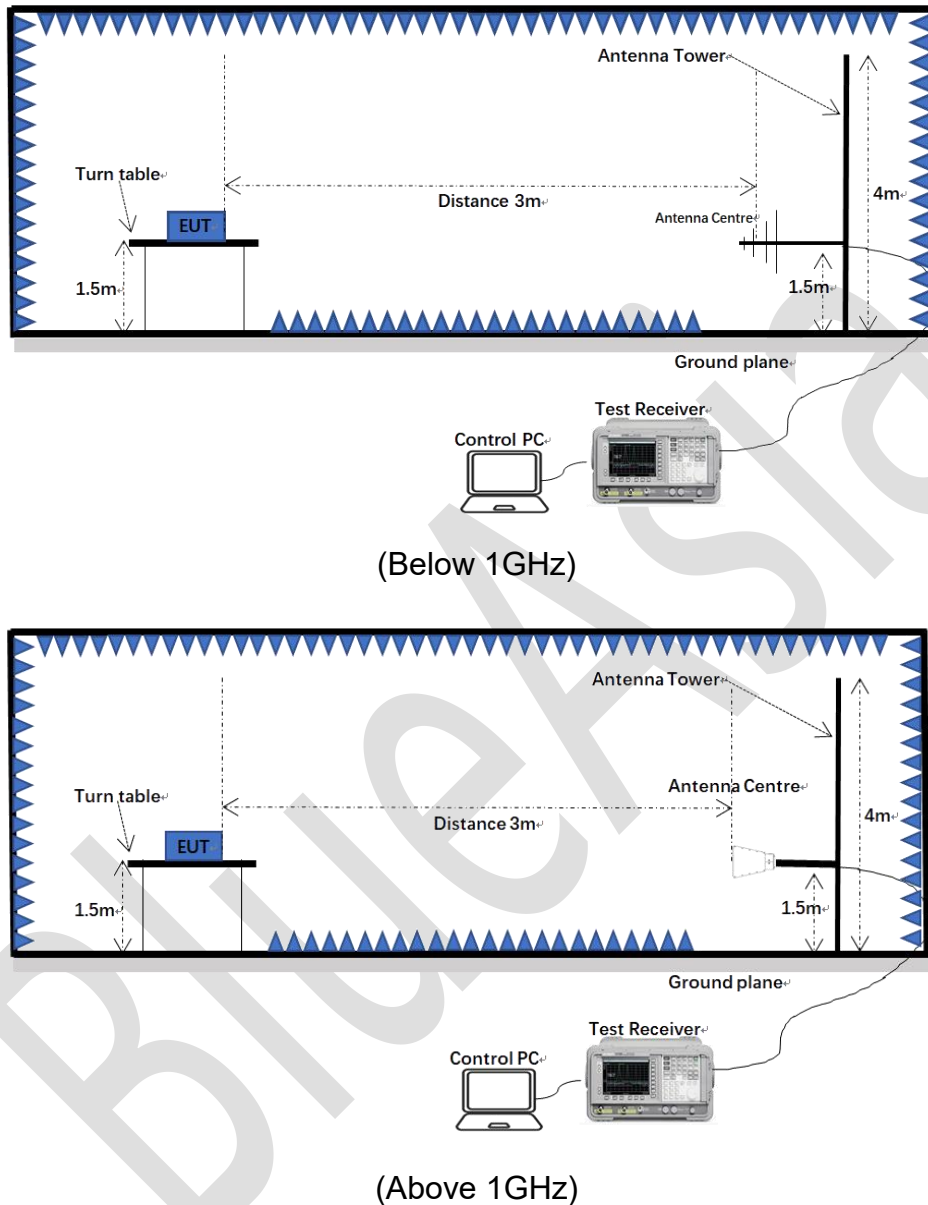
Frequency range	Limit values
$F_{\text{LOWER}} \leq f \leq 1\,000\text{ MHz}$	-57 dBm
$1\,000\text{ MHz} < f \leq F_{\text{UPPER}}$ (see Table 11)	-47 dBm
NOTE: F_{UPPER} and F_{LOWER} are linked with the OFR of the EUT, see Table 11.	

The frequency range for the RX spurious emission test shall be as defined in Table 11.

Table 11: Frequency range for the RX spurious emissions test, linked with EUT OFR according to ERC Recommendation 74-01 [i.15]

Fundamental frequency range defined by f_L and f_H	Frequency range for measurements	
	Lower frequency (F_{LOWER})	Upper frequency (F_{UPPER})
$3,8\text{ GHz} < f \leq 4,2\text{ GHz}$	30 MHz	5 th harmonic (note 1)
$6,0\text{ GHz} < f \leq 8,5\text{ GHz}$	30 MHz	26 GHz
NOTE 1: F_{UPPER} is the stated harmonic of f_H (the upper edge of the OFR, which is measured in clause 4.2.2).		
NOTE 2: F_{LOWER} has to be selected based on f_L and F_{UPPER} based on f_H (f_L and f_H can be measured according to ETSI clause 4.2.2); for receive only devices f_H and f_L of the related EUT/companion device shall be used.		

6.8.2 Test setup



6.8.3 Test procedure

1. Scan from 30MHz to 26GHz; find the maximum radiation frequency to measure.
2. The technique used to find the Spurious Emissions of the transmitter was the antenna substitution method. Substitution method was performed to determine the actual ERP/EIRP emission levels of the EUT.
3. Final Level = Receiver Read level + Correct factor
4. Correct factor = Antenna Factor + Cable Loss – Preamplifier Factor
5. The emission levels of other frequencies are very lower than the limit and not show in test report.

Below 1GHz test procedure as below:

- 1) The EUT was powered on and placed on a table in the chamber. The antenna of the transmitter was extended to its maximum length. Receiver mode and the measuring receiver shall be tuned to the frequency of the transmitter under test.
- 2) Rotating through 360° the turntable. After the fundamental emission was maximized, a field strength measurement was made.
- 3) Steps 1) and 2) were performed with the EUT and the receive antenna in both vertical and horizontal polarization.
- 4) The transmitter was then removed and replaced with another antenna. The center of the antenna was approximately at the same location as the center of the transmitter.
- 5) A signal at the disturbance was fed to the substitution antenna by means of a non-radiating cable. With both the substitution and the receive antennas horizontally polarized, the receive antenna was raised and lowered to obtain a maximum reading at the test receiver. The level of the signal generator was adjusted until the measured field strength level in step 2) is obtained for this set of conditions.
- 6) The output power into the substitution antenna was then measured.
- 7) Steps 5) and 6) were repeated with both antennas vertically polarized.

Above 1GHz test procedure as below:

- 1) Different between above is the test site, change from Semi- Anechoic Chamber to fully Anechoic Chamber.

6.8.4 Test data

6.8.4.1 Below 1GHz

[Test Mode: RX-6500MHz]; [Polarity: Horizontal]



No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector
1	40.5591	-74.92	-9.01	-83.93	-57.00	-26.93	peak
2	78.6888	-64.77	-6.60	-71.37	-57.00	-14.37	peak
3	145.3506	-68.60	-3.58	-72.18	-57.00	-15.18	peak
4 *	157.5588	-66.13	-3.37	-69.50	-57.00	-12.50	peak
5	274.1939	-69.83	-6.17	-76.00	-57.00	-19.00	peak
6	763.3757	-76.58	2.40	-74.18	-57.00	-17.18	peak

Test Result: Pass

[Test Mode: RX-6500MHz] ; [Polarity: Vertical]



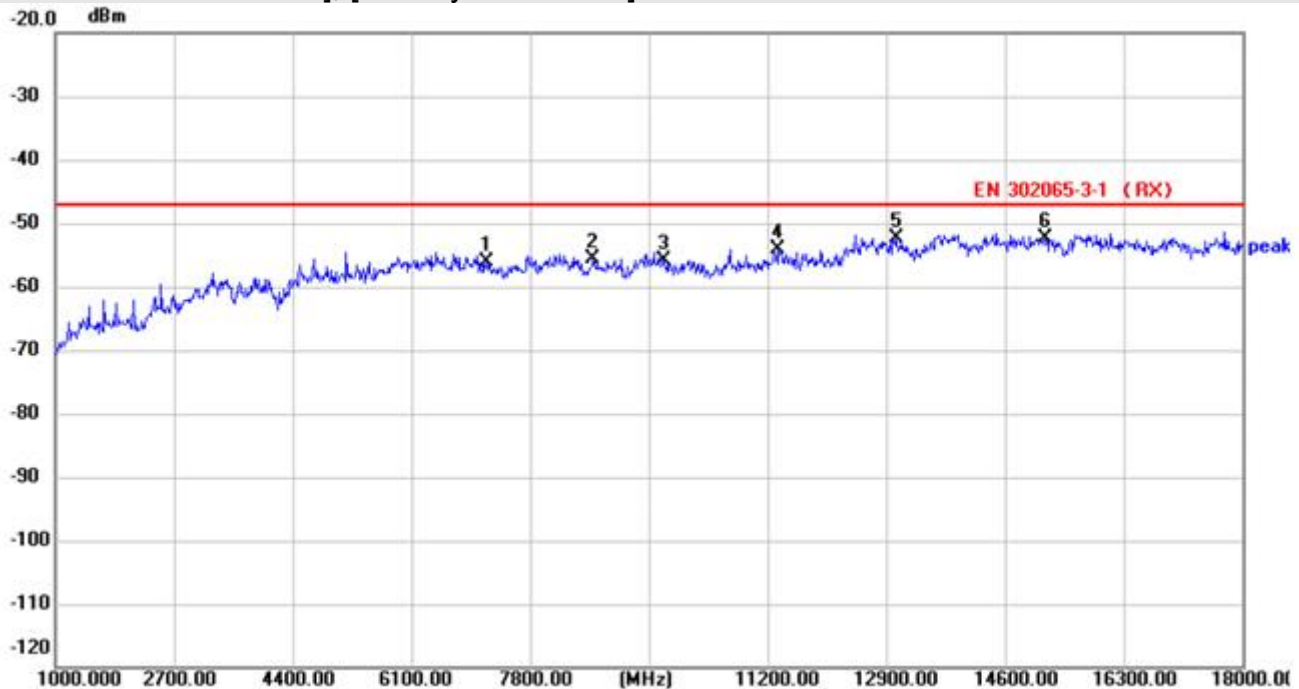
No.	Frequency (MHz)	Reading (dBm)	Factor (dB)	Level (dBm)	Limit (dBm)	Margin (dB)	Detector
1	40.4170	-56.41	-9.04	-65.45	-57.00	-8.45	peak
2	77.0505	-58.25	-6.14	-64.39	-57.00	-7.39	peak
3 *	88.0329	-52.75	-8.30	-61.05	-57.00	-4.05	peak
4	131.7577	-59.09	-4.27	-63.36	-57.00	-6.36	peak
5	160.9089	-63.10	-3.47	-66.57	-57.00	-9.57	peak
6	257.4221	-73.43	-6.67	-80.10	-57.00	-23.10	peak

Test Result: Pass

6.8.4.2 Above 1GHz

Note: The disturbance above 18GHz was very low. The points marked on above plots are the highest emissions could be found when testing, so only above points had been displayed. The amplitude of spurious emissions from the radiator which are attenuated more than 20dB below the limit need not be reported.

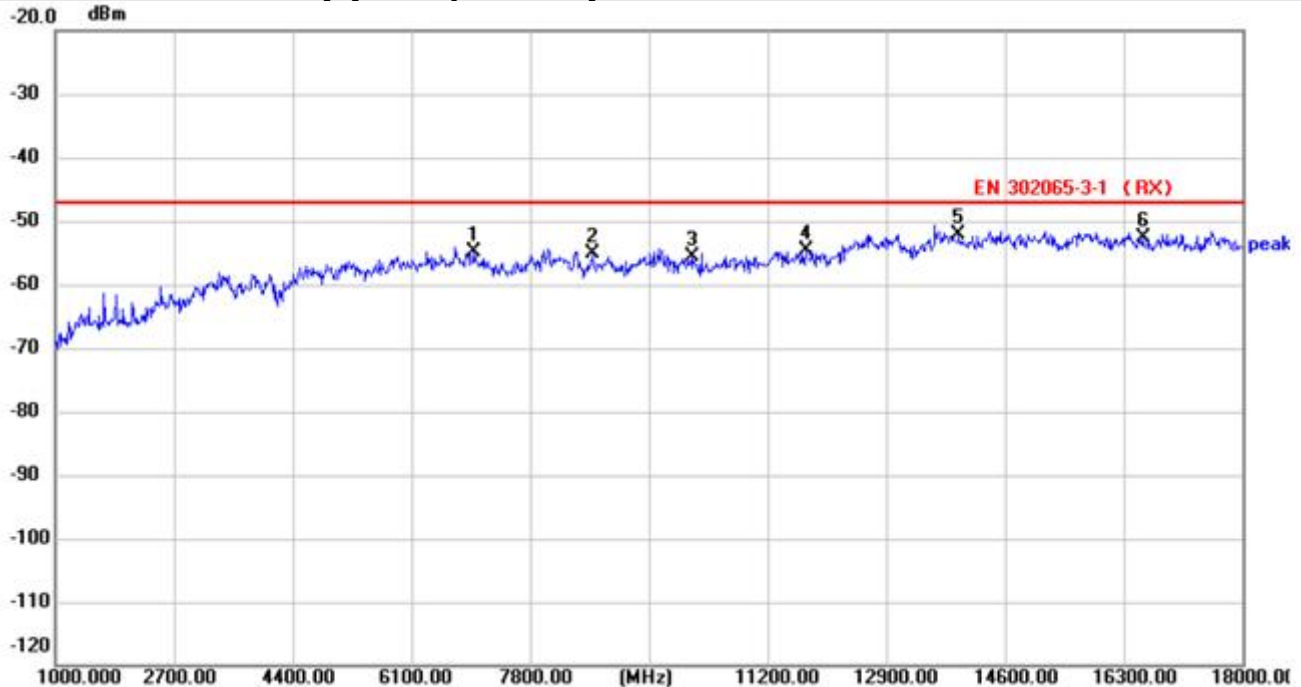
[Test Mode: RX-6500MHz]; [Polarity: Horizontal]



No.	Mk.	Freq. MHz	Reading Level dBm	Correct Factor dB	Measure- ment dBm	Limit dBm	Over dB	Detector
1		7171.000	-69.87	13.78	-56.09	-47.0	-9.09	peak
2		8701.000	-70.40	14.78	-55.62	-47.0	-8.62	peak
3		9704.000	-71.82	15.96	-55.86	-47.0	-8.86	peak
4		11353.00	-71.04	16.86	-54.18	-47.0	-7.18	peak
5	*	13036.00	-73.78	21.43	-52.35	-47.0	-5.35	peak
6		15178.00	-71.13	18.77	-52.36	-47.0	-5.36	peak

Test Result: Pass

[Test Mode: RX-6500MHz]; [Polarity: Vertical]



No.	Mk.	Freq. MHz	Reading Level dBm	Correct Factor dB	Measure- ment dBm	Limit dBm	Over dB	Detector
1		7001.000	-69.09	14.23	-54.86	-47.0	-7.86	peak
2		8684.000	-69.81	14.76	-55.05	-47.0	-8.05	peak
3		10129.00	-72.10	16.40	-55.70	-47.0	-8.70	peak
4		11744.00	-71.63	16.94	-54.69	-47.0	-7.69	peak
5	*	13937.00	-71.44	19.28	-52.16	-47.0	-5.16	peak
6		16572.00	-71.42	18.91	-52.51	-47.0	-5.51	peak

Test Result: Pass

6.9 Receiver Baseline Sensitivity (RBS)

The Receiver Baseline Sensitivity (RBS) requirement applies to all EUT categories as described in Table 2 in clause 4.2.3.

Receiver Baseline Sensitivity is the capability to receive a wanted signal at application related defined input signal levels while providing a pre-determined minimum acceptable level of technical performance.

Two different use cases need to be differentiated due to the different test scenario:

- radio communication applications: EUT (Receiver) and a TX-counterpart; or
- radiodetermination applications: target with defined RCS in a distance to the Transceiver.

6.9.1 Limit

The Receiver Baseline Sensitivity (RBS) is defined as the "received power at the EUT ($P_{@EUT}$)" (according to ETSI EN 303 883-2 [2], clause 5.4.3.3) and shall be at least 70 dB below the maximum Mean e.i.r.p. spectral density level of -41,3 dBm/MHz:

- Receiver Baseline Sensitivity: $P_{@EUT} \leq -111.3 \text{ dBm/MHz}$

NOTE 1: Received power at the EUT ($P_{@EUT}$) includes the RX antenna gain (of the integral or associated or dedicated EUT antenna).

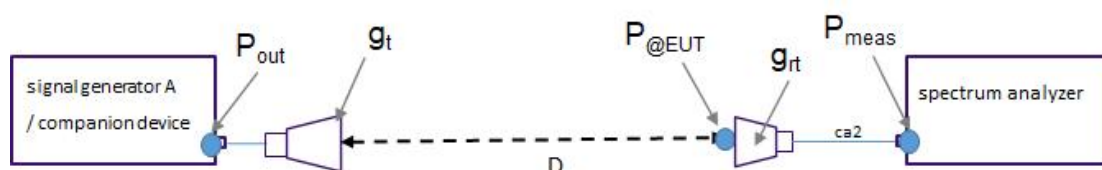
NOTE 2: Reasoning of Baseline Sensitivity requirement:

The Keyless Entry use case (see ETSI TR 103 416 [i.3]) requires reliable operation even under heavy human body attenuation or blockage scenarios. This can be achieved with a good link budget and/or installation of additional UWB nodes at the vehicle. Spectrum efficiency is deteriorated, if link budget due to poor RX sensitivity is traded off against node count, as this will add unnecessary air traffic. This justifies the requirement for a "Baseline Sensitivity".

NOTE 3: Baseline Sensitivity is defined with the same unit as Mean e.i.r.p. spectral density, and therefore the measurement methods for Mean e.i.r.p. spectral density can be reused for quantifying the received power at the EUT.

The wanted technical performance criteria for all Equipment Type 1 and Type 2 for the RBS tests shall be criterion 1, as defined in clause 4.4.2.1.

6.9.2 Test setup



Test procedure

Conformity of Receiver Baseline Sensitivity shall be a radiated test with integral or specified antenna according to ETSI EN 303 883-2 [2], clause 5.4.3.3 ("Radiated measurements for radio communication devices with power limit")

All signal levels (Pout, P@EUT) shall be determined with reference to the Mean e.i.r.p. spectral density and therefore documented with the unit "dBm/MHz".

The wanted technical performance criteria for the test are provided in clause 4.4.2.1. of ETSI EN 302 065-3-1 V3.2.1 (2025-02)

6.9.3 Test data

Test Frequency (MHz)	Wanted signal (dBm)	Factor at 3m	Mesurment P@EUT	MSR	Limit of RBS (dBm/MHz)	Result
6500	-104.1	14.20	-118.3	100	-111.3	Pass
Note1: MSR mean message success rate						
Note2: The MSR shall be determined with 20 messages.						

6.10 Receiver Baseline Resilience (RBR)

The Receiver Baseline Resilience (RBR) requirement applies to all EUT categories as described in Table 2 in clause 4.2.3. of ETSI EN 302 065-3-1 V3.2.1 (2025-02)

For the description of the Receiver Baseline Resilience (RBR), see ETSI EN 303 883-2 [2], clause 5.5.

6.10.1 Limit

The present document considers two interference scenarios:

- Interference Scenario 1: Weak interference: Wanted Technical Performance Criterion 1 (clause 4.4.2.1) shall be fulfilled.
- Interference scenario 2: Strong interference: Wanted Technical Performance Criterion 2 (clause 4.4.2.2) shall be fulfilled. Both interference scenarios shall be tested and the EUT shall comply with the limits given in Table 12 and Table 13.

Limits for the interferer within OFR shall be as shown in Table 12.

Table 12: RBR limits within OFR

	Interference Scenario	Interference power level at EUT	Wanted technical performance criterion	Test frequencies	Modulation of test signals
Type 1 equipment	Scenario 1	-85 dBm	Criterion 1	f_L (lower edge of OFR) f_c (centre frequency of wanted signal) f_H (upper edge of OFR)	CW
	Scenario 2	-50 dBm	Criterion 2		
Type 2 equipment	Scenario 1	-75 dBm	Criterion 1		
	Scenario 2	-50 dBm	Criterion 2		

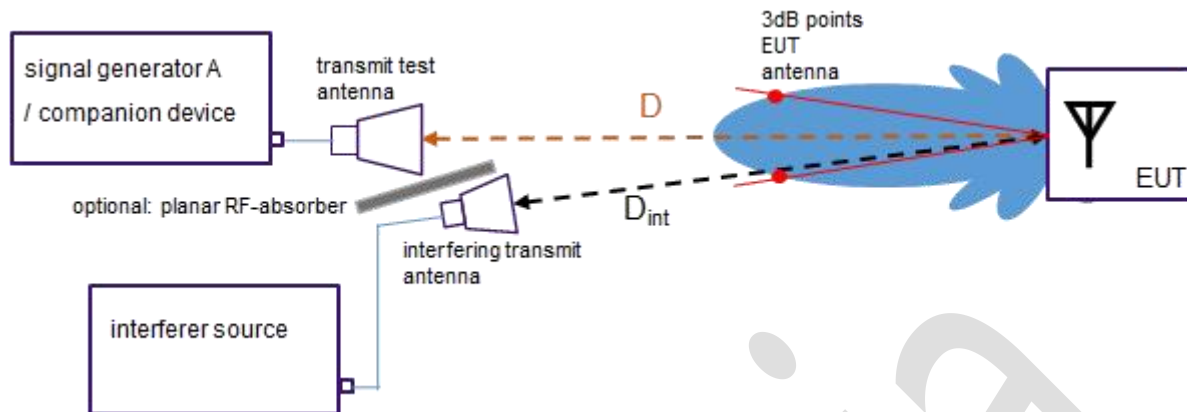
Limits for the interferer outside OFR shall be as shown in Table 13.

Table 13: RBR limits outside OFR

	Interference Scenario	Interference power level at EUT	Wanted technical performance criterion	Test frequencies	Modulation of test signals
Type 1 equipment	Scenario 1	-85 dBm	Criterion 1	$f_c - 2 \times \text{OFR}$ $f_c - 1 \times \text{OFR}$ $f_c + 1 \times \text{OFR}$ $f_c + 2 \times \text{OFR}$	CW
	Scenario 2	-50 dBm	Criterion 2		
Type 2 equipment	Scenario 1	-75 dBm	Criterion 1		
	Scenario 2	-50 dBm	Criterion 2		

NOTE: The derivation of the above limits in Tables 12 and 13 is described in annex D.

6.10.2 Test setup



6.10.3 Test procedure

Conformity of Receiver Baseline Resilience shall be tested according to ETSI EN 303 883-2 [2], clause 5.5.3.3 ("Radiated Measurements for Radio Communication Devices with Power Limit") with the following parameters:

The wanted technical performance criteria for the test are provided in clauses 4.4.2.1 and 4.4.2.2. of ETSI EN 302 065-3-1 V3.2.1 (2025-02)

The wanted signal level for all RBR tests shall be 10 dB above the Receiver Baseline Sensitivity (see clause 4.4.4.3) of ETSI EN 302 065-3-1 V3.2.1 (2025-02).

Sensitivity degradation $d_g = 10$ dB

The wanted signal shall be at its nominal frequency f_c .

Interfering signals within OFR are given in Table 12 and Interfering signals outside OFR in Table 13 of ETSI EN 302 065-3-1 V3.2.1 (2025-02)

For interference scenario 2 the interference shall be present for 10 seconds. After 11 seconds the MSR measurement is started again and the EUT shall fulfil message success rate better than 90 %.

6.10.4 Test data

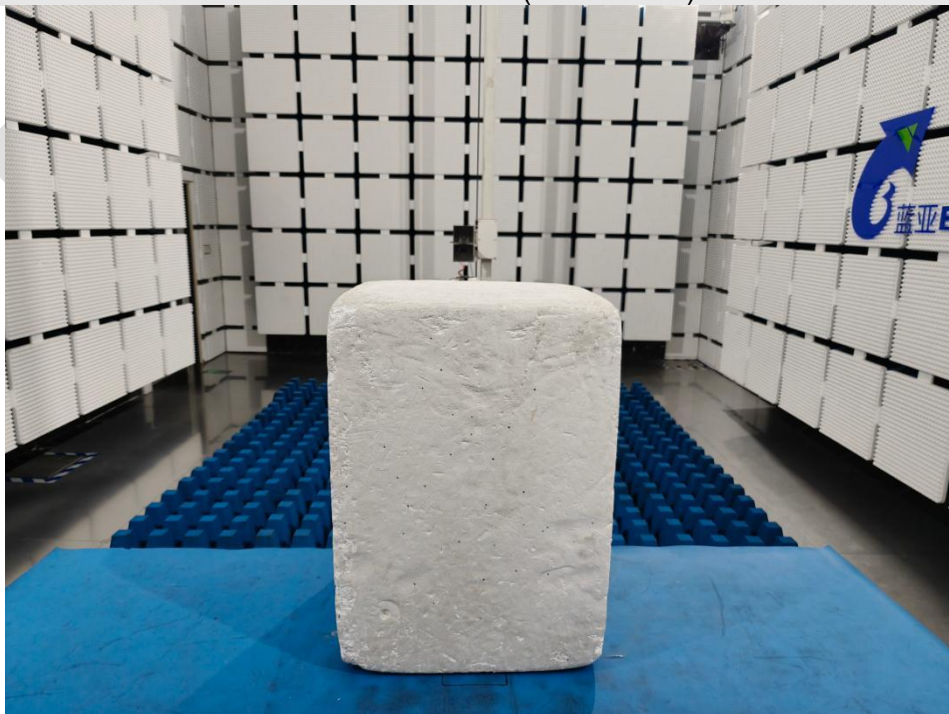
Interference Scenario	Test Frequency (MHz)	Wanted signal (dBm)	Factor at 3m	Interference frequency	Interference signal (dBm) CW	Mesurment $P_{@EUT}$	MSR	Limit of RBS (dBm/MHz)	Result
Weak interference	6500	-103.2	14.2	$f_c - 2 \times OFR$	-75	-117.40	100	-111.3	Pass
		-103.1	14.2	$f_c - 1 \times OFR$	-75	-117.30	95	-111.3	Pass
		-105.2	14.2	OFR	-75	-119.40	100	-111.3	Pass
		-105.0	14.2	$f_c + 1 \times OFR$	-75	-119.20	95	-111.3	Pass
		-103.3	14.2	$f_c + 2 \times OFR$	-75	-117.50	100	-111.3	Pass
Strong interference	6500	-104.4	14.2	$f_c - 2 \times OFR$	-50	-118.60	95	-111.3	Pass
		-104.1	14.2	$f_c - 1 \times OFR$	-50	-118.30	100	-111.3	Pass
		-105.4	14.2	OFR	-50	-119.60	100	-111.3	Pass
		-105.3	14.2	$f_c + 1 \times OFR$	-50	-119.50	100	-111.3	Pass
		-104.2	14.2	$f_c + 2 \times OFR$	-50	-118.40	100	-111.3	Pass
Note1: MSR mean message success rate Note2:The MSR shall be determined with 20 messages.									

7 Appendix A: photographs of test setup

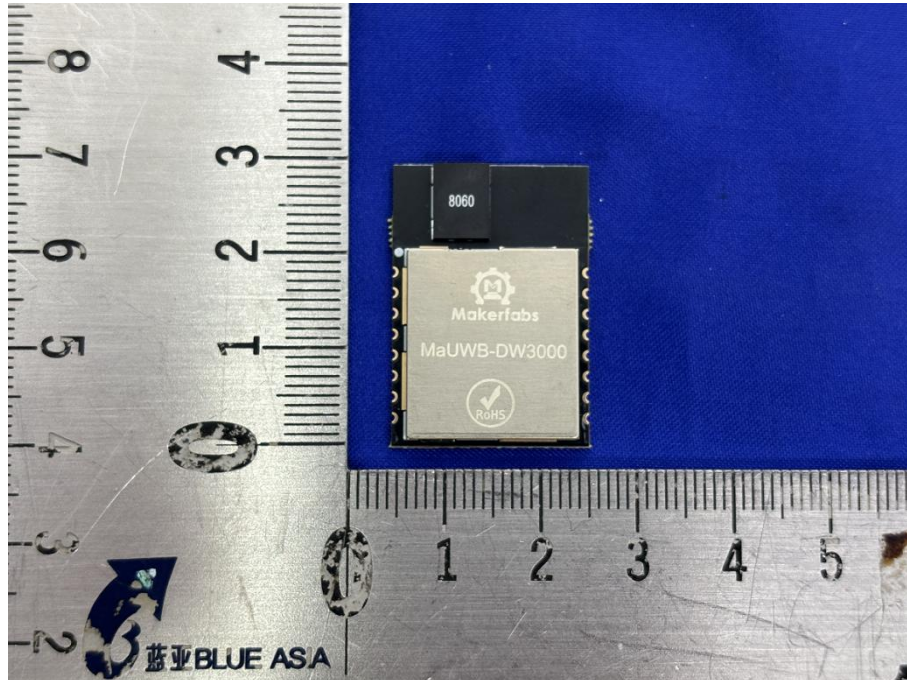
Radiated Emissions (30MHz-1GHz)



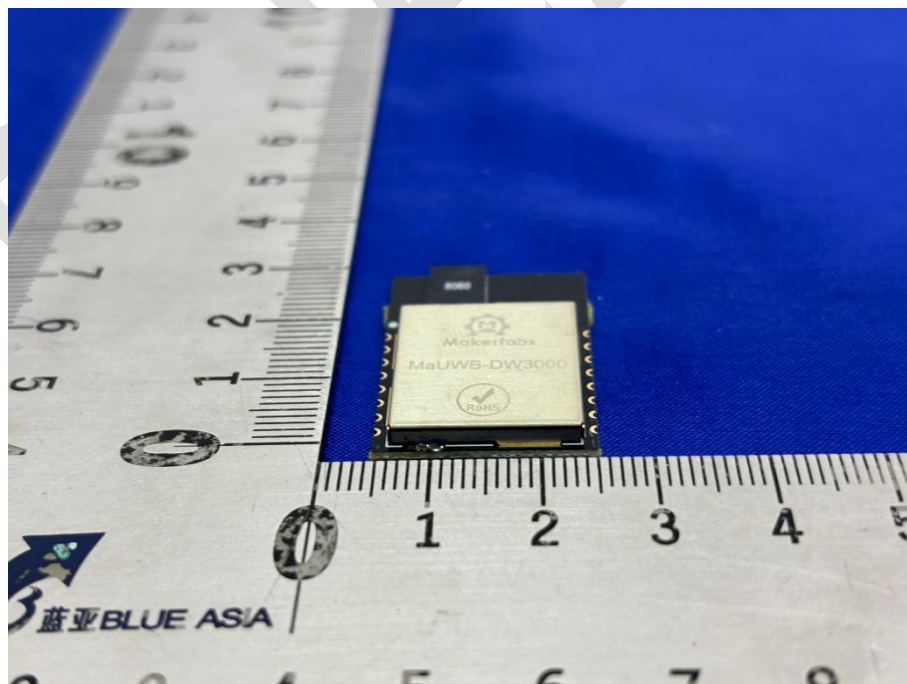
Radiated Emissions (above 1GHz)



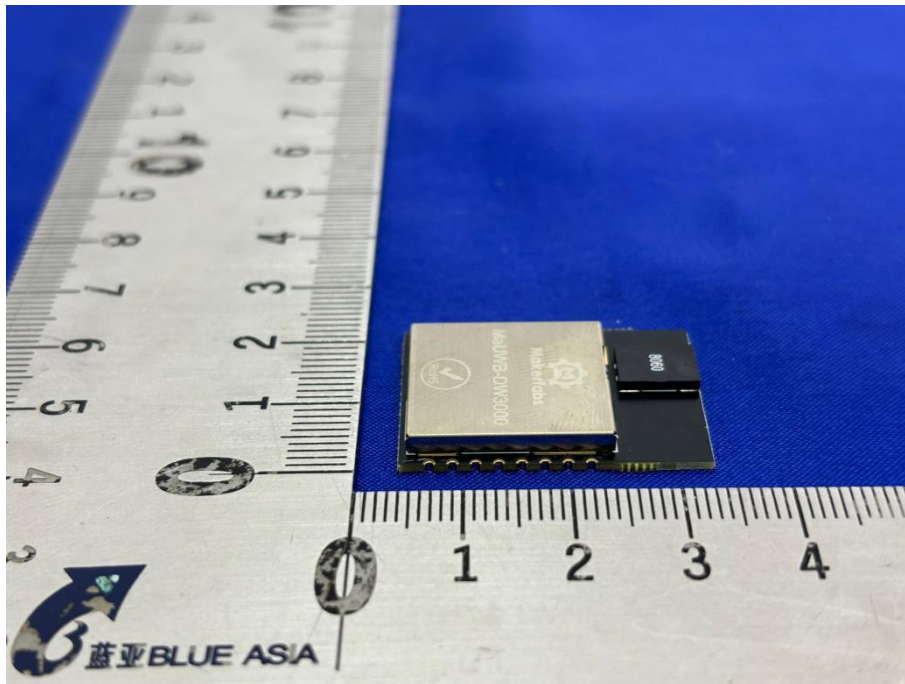
Appendix B: photographs of EUT



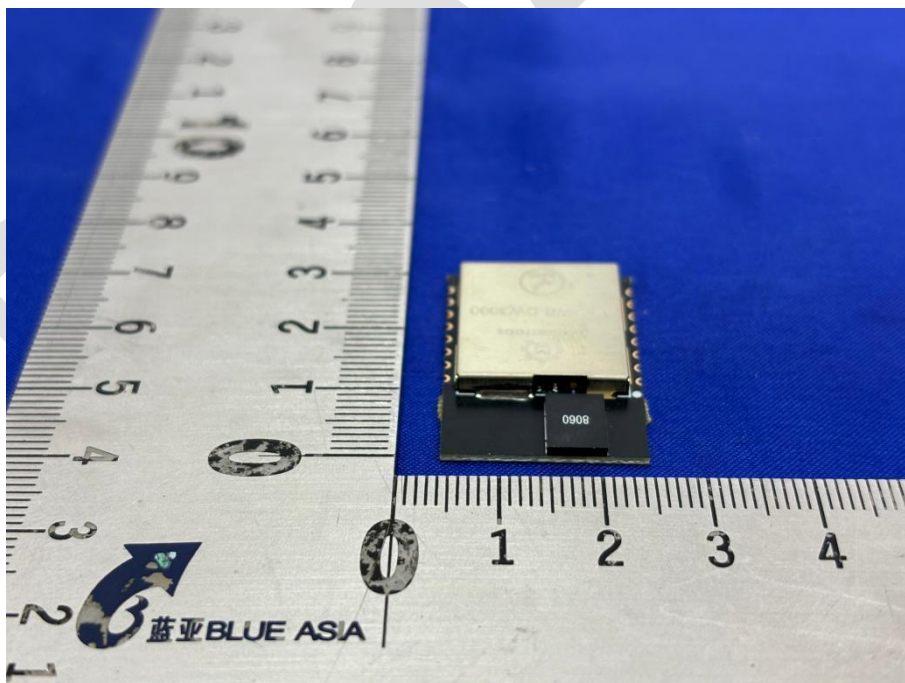
View of Product-1



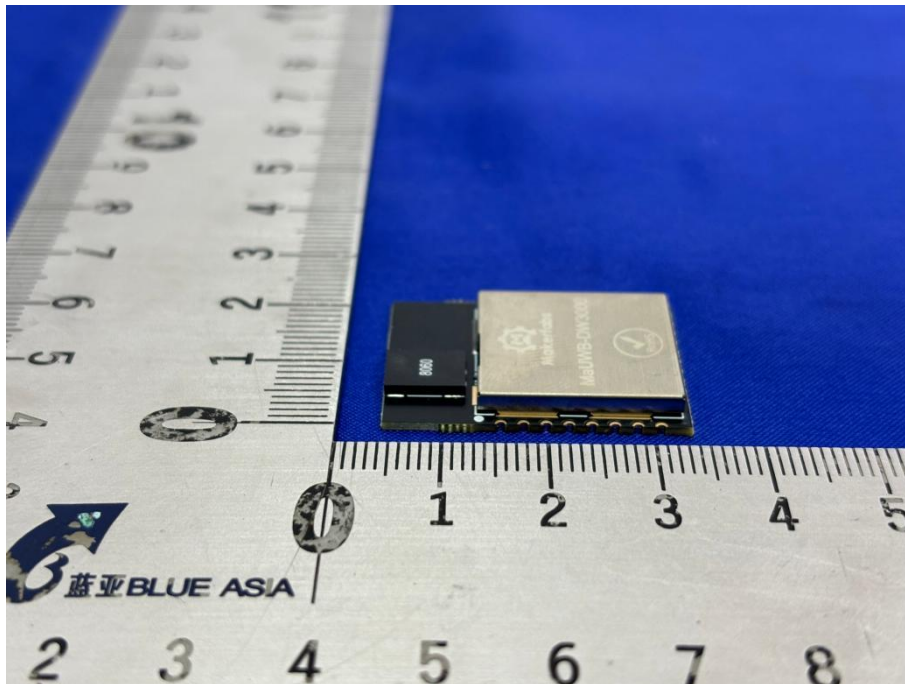
View of Product-2



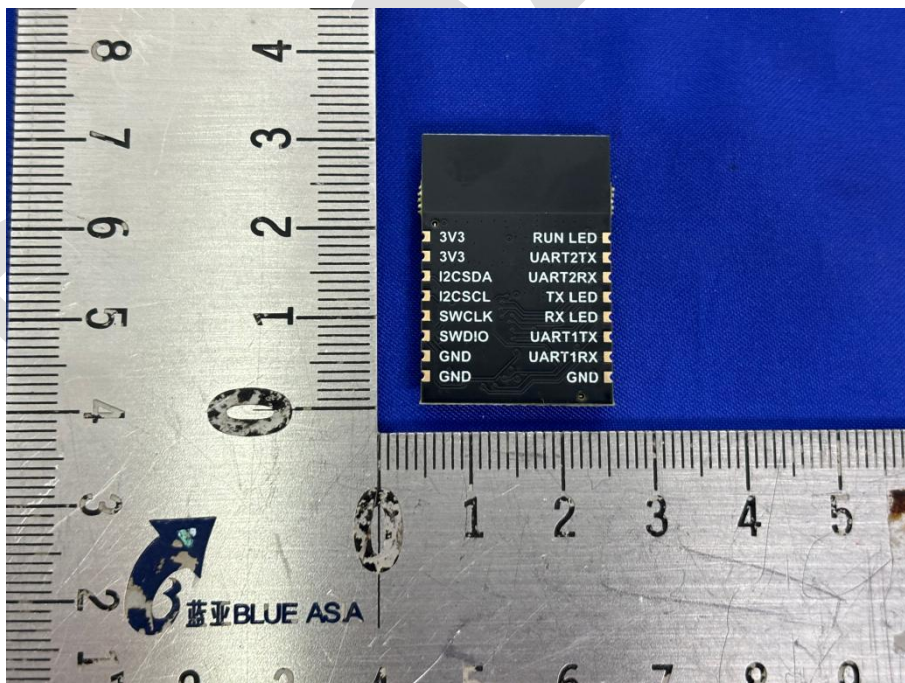
View of Product-3



View of Product-4



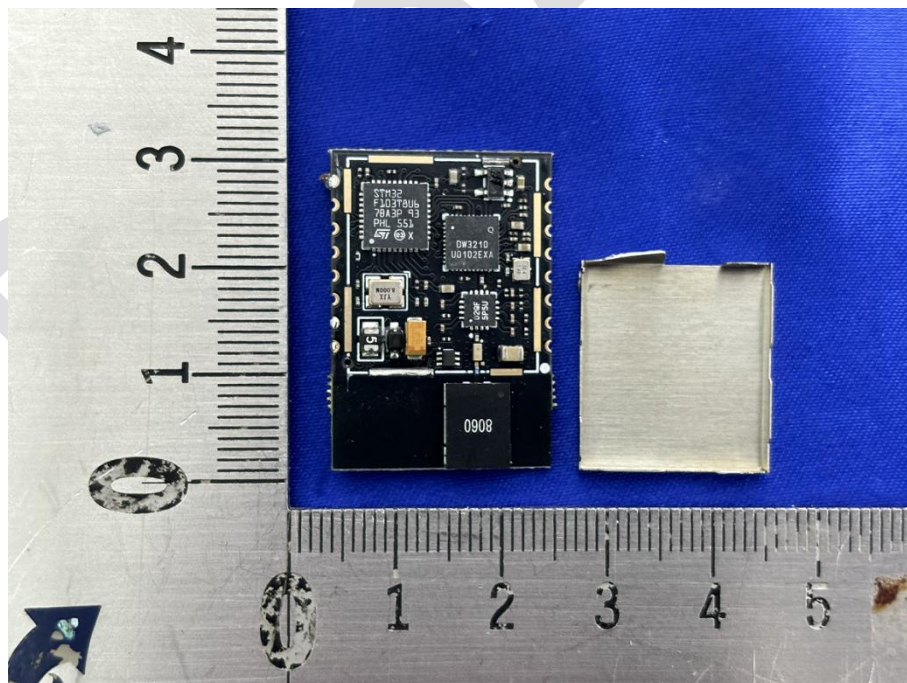
View of Product-5



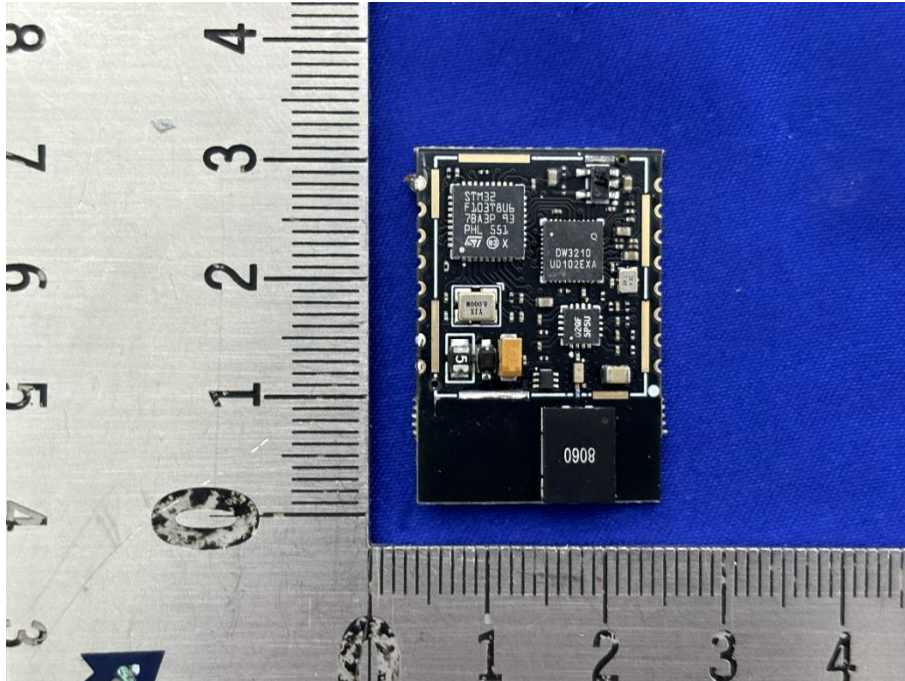
View of Product-6



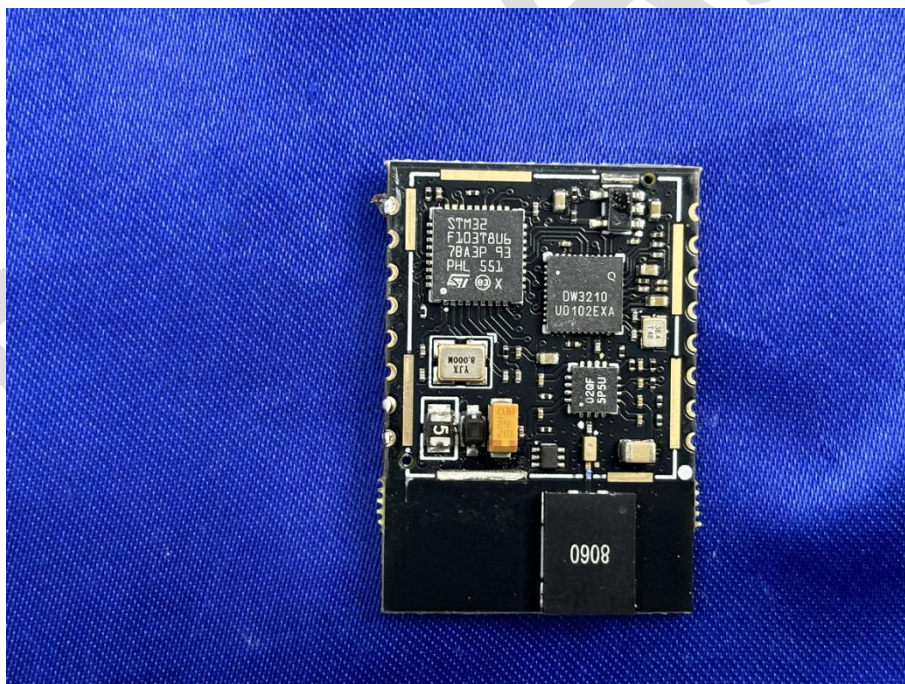
View of Product-7



View of Product-8



View of Product-9



View of Product-10

----END OF REPORT----

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